

Manual

Inspection Planning of Initial Sample Inspections FEP-PPE 8.1

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1 Inspection Planning of Initial Sample Inspections

Purpose

This component makes it possible to create inspection plans based on master data that was set up previously and to create/ manage inspection requirements in order to prepare the documentation for initial sample inspection.

Implementation considerations

Because the prepared inspection plans form the basis of the initial sample inspection that is conducted later, this component is required if measured values and attributive inspection decisions are to be recorded during the initial sample inspection.

Integration

The element is primarily used to control the components.

- Collection/ info function for quality data (AIP)
- Expanded inspection planning/ inspection steps
- Family inspection planning
- Reports of in-production inspections and the
- Failure mode analysis/ measures tracking

Features

The following functions are available:

- Update function to enter and edit relevant master data (articles, companies, defects, measures, characteristics, etc.)
- Update function for creating and modifying inspection plans
- Versioning of inspection plans with history administration
- Function for releasing and activating inspection plans using separate rights
- Function for attributing or modifying inspection plan characteristics for various characteristic types (variable, attributive)
- Function for assigning operations and inspection stations used to structure the inspection steps to be generated based on the inspection plan
- The ability to define tolerance limits, the unit, the sampling scheme, the control chart to be used, activation of the automatic defect generation, etc.
- The ability to assign any document (drawings, images, videos of any formats as well as internal notes) to characteristics, inspection plans, inspection plan characteristics, inspection requirements or inspection step characteristics in a list of documents.

- Function to generate inspection requirements/ inspections steps based on inspections plans that were previously created

2 Summary

2.1 Notes on the Document

This document describes the “article” application of the Manufacturing Operation Center (MOC). General information on how to use MOC can be found in the document entitled “[moc_cc.pdf](#)”.

3 Article

The article catalog has been designed to edit/keep articles. Article data is a global catalog that is used in many CAQ modules and in PDV (Process Data Collection). Provided that there is an interface to a higher-level system (e.g. ERP system), articles may be created automatically via this interface. As soon as a new article is created or changed, e.g. in the ERP system, the article data record is automatically created or changed in the HYDRA-CAQ article catalog based on the defined information.

3.1 Starting the Function

Menu	Master data → Quality management → Article Master data → Process data processing → Article
Transaction code	atc
Function authorization	atc

3.2 Default Application Layout

Id	Drawing issue nu...	Designation	Article customer num...	Article
23 9383 712	B	Headlight XA 77		VW
32 9382 111	G	Washer M5		
77 9938 882	C	Metal Pen, 4-colour, 10000m capacity	MKS 234-2	Giant
77 9938 885	D	Plastic Pen, single colour, 6000m capacity		LOGI
88 9938 883	B	Metal Pen, single colour, 10000m capacity		Giant
ET-8440-A		granules		
SAL-FFI-9005		mirror left black		Ford

3.3 Toolbar

The toolbar provides the different functions available for this application and possibly links to other applications. The functions included in the “general” tab of the toolbar refer to all detail applications. In addition to the standard functions, such as help, request data, save application settings, and print preview, the other tabs also include specific functions that are specially tailored to the respective detail application. The following sections describe the individual application functions.

“Data” Category



Request data

The information to be displayed within the application is requested on the basis of the entered selection criteria. This process might take some time depending on the dataset from which the system filters data and on the selection result to be transferred and displayed.



Cancel

The query sent by clicking the “request data” button can be canceled using this function.

**Print preview**

The print preview is opened for the selected detail application. The print preview also includes further options to change the resulting printout and functions for exporting the displayed information into other formats, such as PDF, Excel, image files.

**Save**

The application design configured by the user, e.g. columns and categories displayed as well as their respective size and display locations, etc. are only saved if the user requests it. In this case, the user has to affirm the confirmation prompt by clicking "Yes".

"Functions" Category**Add**

Adds a new article.

**Copy**

Copies the selected article.

**Edit**

Edits an already existing article

**Delete**

Deletes the selected or several selected articles.

"Help" Category**Help on operation**

Clicking this button opens the help file describing how to operate MOC. The basic document is entitled "[moc_cc.pdf](#)". It describes how to use MOC in general and applies for all applications.

**Help on application**

This function opens the manual corresponding to the respective application from which the help was requested. The application manual integrates the application function into the MES context and explains the information to be displayed. The documentation also includes all detailed applications.



Help on detail application

This function opens the application manual at the section where the respective detailed application is described.

3.4 Selection parameters

The application provides the following selection criteria:

“General“ tab

- Article no.:
Article number
- Drawing issue number:
Drawing issue number of the article, often also referred to as index
- Designation:
Article name
- Inactive:
Inactive, active articles. The checkbox is not activated by default.
- Customer article no.:
Customer article number
- Article model:
Article model

“Groups“ tab

- Group:
The article group tree can be opened using the  button if an article group is to be filtered. There is a function to accept and cancel the activity.

3.5 “Article” Detail Application

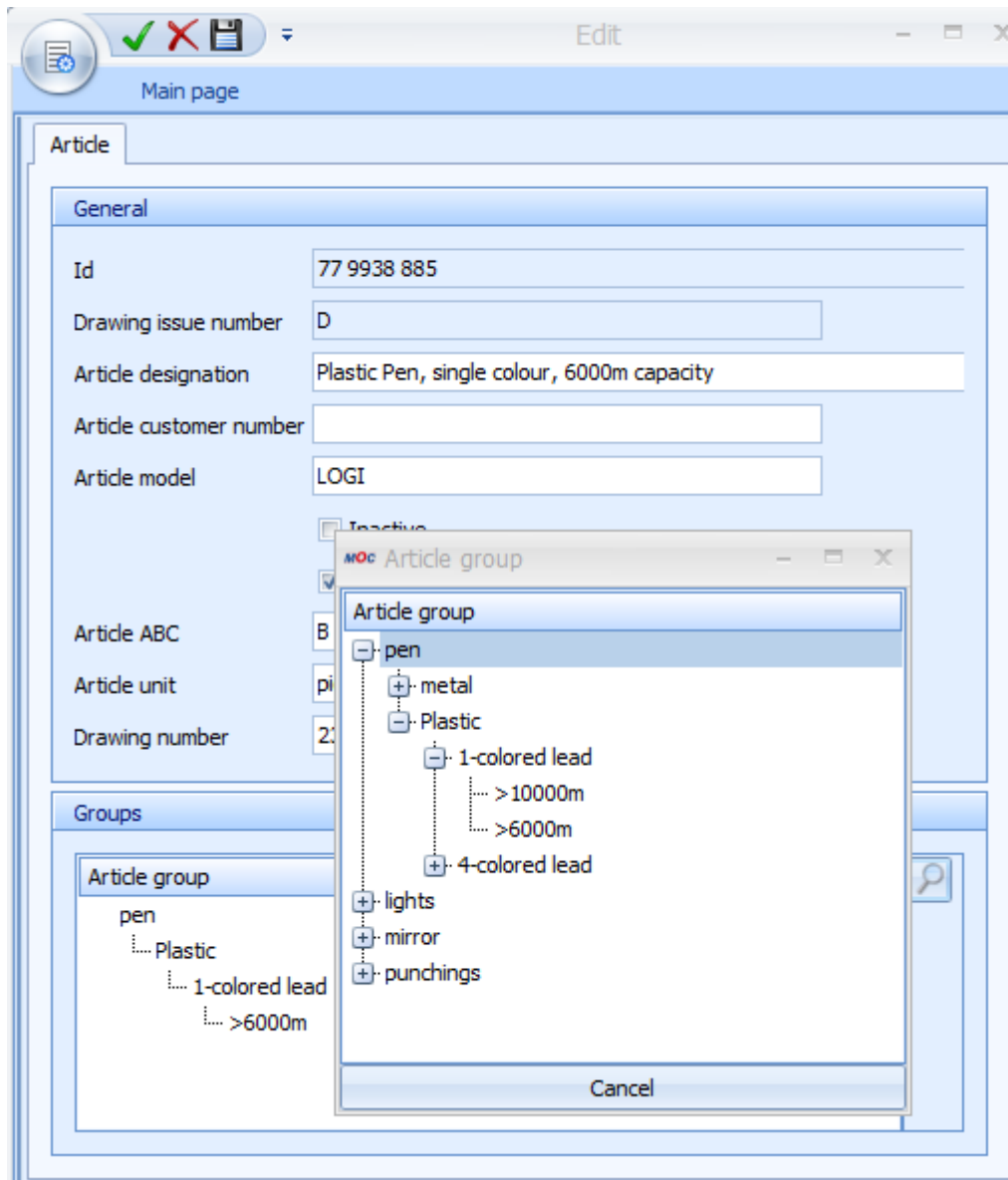
The article number as well as the drawing issue number uniquely identify articles in all areas of HYDRA-CAQ referring to the article catalog. The drawing issue number, also referred to as article index, may be very important, in particular, for inspection planning and when inspection orders are generated. Thus, it is possible to create, for example, an inspection plan for article 12938 with the drawing issue number A and B within inspection planning, whereas there are different inspection specifications for each drawing issue number. Unless the drawing issue number is indicated and thus may be part of the inspection plan, the system that generates the inspection requirements, must deliver this drawing issue number.

Consequently, the “article no.” and “drawing issue number” fields are key fields, i.e. when a new article is saved, it is first checked whether an article with this key information already exists.

By distinguishing between active and inactive articles, it may be defined whether or not the articles are available in certain selection lists. Thus, no inspection plan can be created for an inactive article. However, inactive articles may be evaluated at any time. Moreover, inactive articles can also be reactivated at any time.

Furthermore, an article can be defined as being subject to documentation. In addition the dialog provides the fields customer article number, article model, article ABC, drawing number as well as the possibility to assign units. The catalog of units is accessed to assign units (dimensions).

The corresponding group needs to be assigned, provided that article groups are to be evaluated or family inspection plans are to be used at a later point in time. Groups are assigned by opening the group tree using the magnifier button. Using the hierarchic tree entries the required group can be selected in the group tree and accepted by double clicking.



The assigned group including the hierarchical group structure then appears in the “groups” field of the editing dialog of articles.

When articles are displayed in a list, the group hierarchy is represented by the columns “group 1 to group 5”.

Groups are maintained in the “article groups” application and are described in the document entitled “[MOC_Groups.pdf](#)”.

4 Summary

4.1 General notes on the document

This document describes the “Groups“, e.g. article groups, application of the Manufacturing Operation Center (MOC). For general information on how to use MOC, please refer to the “[moc_cc.pdf](#)“ document.

5 Groups

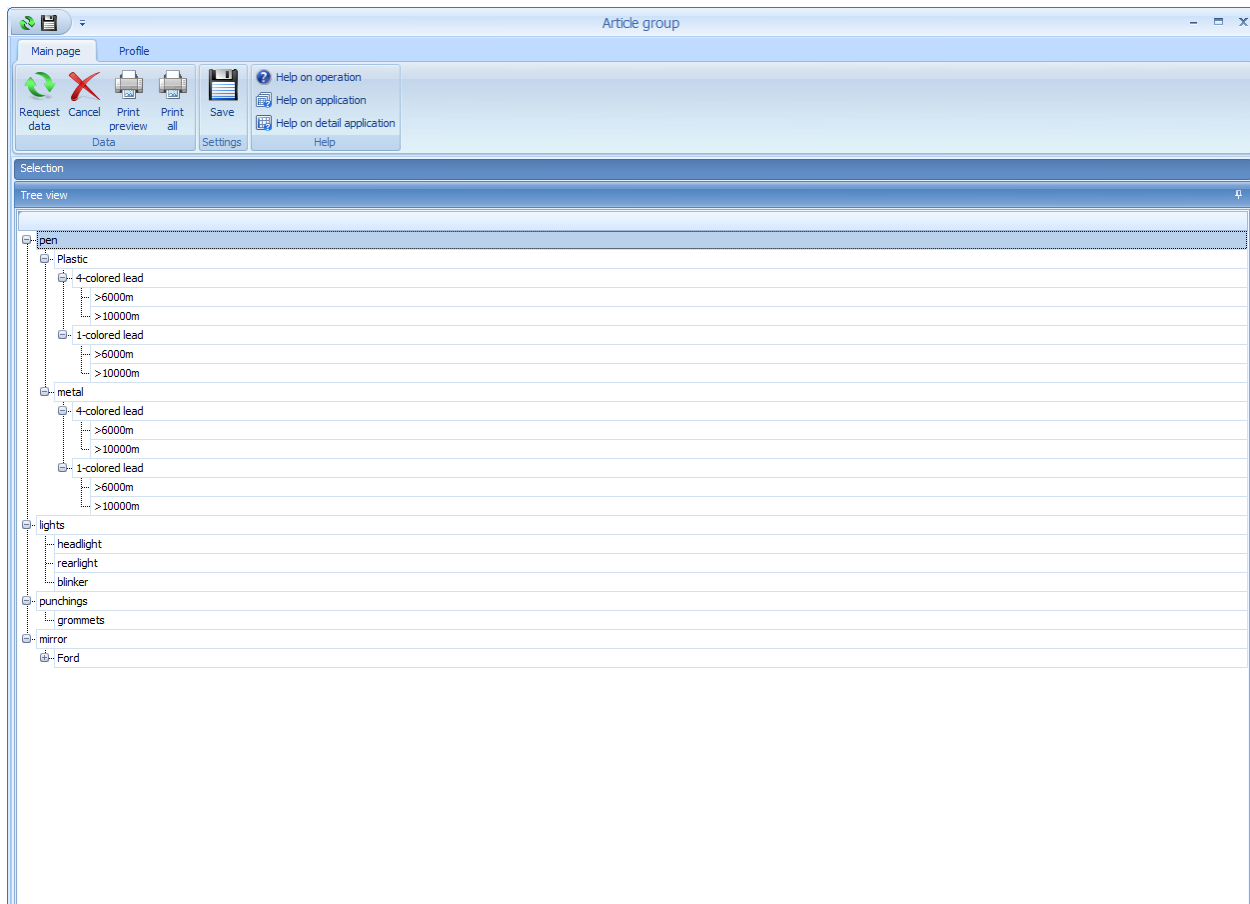
The group catalogs have been designed to create and edit groups for the different applications. The created groups may be assigned to master data of the corresponding application. Consequently, article groups may be created, for example, and assigned to the articles. In this case, it is also possible to create inspection plans on the basis of article groups.

Basically, the creation of groups is also reasonable for failure mode analyses.

5.1 Starting the function

Menu	Master data → Quality management → Article groups Master data → Process data processing → Article groups Master data → Quality management → Measure groups Master data → Quality management → Failure type groups Master data → Quality management → Failure location groups Master data → Quality management → Failure cause groups Master data → Quality management → Causer groups Master data → Quality management → Cost type groups
Transaction code	atcgr → Article groups measgr → Measure groups ftypgr → Failure type groups flocgr → Failure location groups fcaus → Failure cause groups origr → Causer groups costgr → Cost type groups
Function authorization	atcgr -> Article groups measgr.* → Measure groups ftypgr.* → Failure type groups flocgr.* → Failure location groups fcaugr.* → Failure cause groups origr.* → Causer groups costgr.* → Cost type groups

5.2 Default Application Layout



This figure of the article group catalog is exemplary for all groups.

5.3 Toolbar

The toolbar contains the function calls that are available for this application and possibly links to other applications. The functions placed on the "general" tab of the toolbar refer to all detail applications. In addition to the standard functions such as "help", "request data", "save application settings" and "print preview", the other tabs include specific functions that are tailored to the corresponding detail application. The individual functions of the application are listed in the paragraphs that follow.

"Data" category



Request data

The information to be displayed within the application is requested according to the entered selection criteria. This process might take some time depending on the data set from which the system filters data and on the selection result to be transferred and displayed.



Cancel

This function cancels the query sent by clicking the "request data" button.



Print

preview

The print preview is opened for the selected detail application. The print preview also includes further options to change the resulting printout and functions for exporting the displayed information into other formats, such as PDF, Excel, image files.



Save

The application design configured by the user, e.g. columns and categories displayed as well as their respective size and display locations, etc. are only saved if the user requests it. In this case, the user has to affirm the confirmation prompt by clicking "Yes".

"Functions" category

There are no special functions for this detail application. Groups are created, changed and deleted using the context menu of the right mouse button within the tree structure view.

"Help" category



Help on operation

Clicking this button opens the help file describing how to operate MOC. The basic document is entitled "moc-cc.pdf". It describes how to use MOC in general and applies for all applications.



Help on application

This function opens the manual that describes the application from which the help function was requested. The application manual integrates the application function into the MES context and explains the information to be displayed. The documentation also includes all detailed applications.



Help on detail application

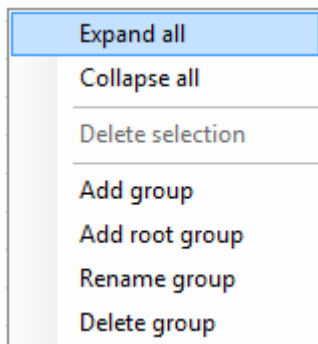
This function opens the application manual at the section where the respective detailed application is described.

5.4 Selection parameters

There are no selection parameters. A specific group can be found by using the “fast selection” function. To use the “fast selection” function just open the group tree structure on the 1st level, select the first entry and enter the first letter of the group in question. Consequently, the first group starting with this letter that is found is selected. The “fast selection” function also integrates subordinate groups that are not opened. If the requested term is included in a group that is not opened, it will be opened automatically.

5.5 "Groups" Detail Application

A group may be created, changed or deleted by opening the context menu of the right mouse button within the display area of the tree structure.



It is altogether possible to define groups up to the fifth hierarchy level. The “add root group” function has to be selected in the context menu of the group tree to create a new main group (1st level). The menu entry “add group” generates a sub-group (level 2 to 5). A designation, which is directly entered in the list view, has to be assigned for this new group. To be able to save the new group, click above or below this new entry within the group tree. Then a confirmation prompt appears asking whether or not the new group is to be saved. Provided that this question is affirmed (“yes”), the entry is saved. The same applies for renaming of groups. Regardless of which hierarchy level is concerned, the entry has to be selected to be able to edit the group designation. Changes can directly be entered in the corresponding line of the tree view. Click above or below the entry to be changed to be able to save the modification. No confirmation prompt appears when it comes to renaming.

An entry is also deleted by selecting a group entry and executing the “delete group” function. Only the group that is at the bottom of the group tree can be deleted.

The “expand all” context menu entry opens all groups up to the lowest hierarchy level. The “collapse all” context menu option closes all entries up to the first level.

The "delete selection" function cannot be used in the maintenance of groups dialog. This function is enabled, for example, in the maintenance of articles application if an article group is selected and this selection is to be removed/deleted.

6 Characteristic Master Data

Summary

Menu	Master Data → Quality Management → Characteristics
Transaction code	chrq
Function authorization	chrq

Utilization

The characteristics catalog has been designed to define characteristics and, as a result, to predefine characteristic data of inspection plans. For this reason, it is directed to people involved in inspection planning.

The characteristics catalog is one of the most important basic catalogs for without this catalog no inspection plan could be created. As it has been designed to predefine characteristics data for inspection plans it provides extensive input possibilities. Generally, only this data, which has not to be modified at a later point in time when assigning it in the inspection plan, should be entered in the characteristics catalog. The definition of limit values, for example, is usually not reasonable as they are only known when the inspection plan is created. For a relation to a specific article is only established with the assignment to an inspection plan. If you keep this always in mind it becomes obvious which information needs to be defined. For this reason, it should be considered thoroughly whether the characteristic "outer diameter" is to be created only once and the detailed information is to be defined later in inspection planning or whether it would be better to create several characteristics for the "outer diameter" specifying the limit values already. In the majority of cases, it is better to only enter a few general characteristics. The required evaluations/reports also play a role in this. If a new "outer diameter characteristic" is created for nearly every tolerance modification it is almost only "applicable" for one article and comprehensive evaluations are not possible when failures are analyzed later"

It is important that each detail defined here can be modified in the inspection plan or that details, which have not been stated, can be added.

The configurations made in the characteristic master data are not final. Characteristic master data has rather been designed as a template for inspection planning. All settings of the characteristic master data can be supplemented and modified in inspection planning.

Integration

The characteristic catalog is a global catalog that is used in many QM applications. Please find below some possible fields of application that refer to the characteristic catalog.

- Inspection planning for production, goods receipt, goods issue, initial samples and calibration
- Inspection requirements for production, goods receipt, goods issue, initial samples and calibration
- Failure analysis in complaints management
- Several reports/evaluations

Prerequisite

There are no special requirements.

Selection criteria

The application provides the following selection criteria:

- Characteristic no.:
Characteristics number
- Characteristic designation:
Designation of the characteristic – Please note: wildcards may be used "*"
- Characteristic type:
Inspection type: attributive, inspection chart, variable

"Details" tab:

- Gages

Selects a gage
- Gage designation:
Selects a gage designation

"User fields" tab:

- If user fields are created they may be selected

If several selection criteria are used overlapping results are displayed in the characteristic master data.

In addition, the column filter allows for the content of each individual column to be filtered.

Field descriptions

The available fields are self-explanatory and are not explained separately, except for the address fields.

"Characteristics" tab

Characteristic no.

Unique number of the characteristic

Characteristic designation

Designation of the characteristic

Input type

Automatic or manual collection type. This field controls the release of HYDRA-PDV fields (for automatic collection). If the automatic collection function is selected, the characteristic type is restricted to the "variable" option.

Characteristic type

Decides whether the inspection is based on the collection of measured values (variable) or on the indication of the number of identified errors (attributive). When it comes to the attributive inspection, the decision is often based on "pass" or "fail". The inspection chart and the information characteristic are further characteristic types. The information characteristic has only been designed to display a document while the inspection is running. Subject to the input type, the lower area of the dialog provides corresponding sampling schemes.

Inspection result base

Decides whether all samples or only the last sample are used to determine the inspection result (pass/fail).

Mandatory inspection

If this checkbox is activated, at least one measured value has to be entered for this characteristic, before an inspection order can be completed with this characteristic.

Calculate characteristic

Identifies a characteristic to be calculated

The authorization "iriscp.formula" is required to show this field in inspection plan characteristics.

Formula:

Please also see the section entitled "formula calculation"

The authorization "iriscp.formula" is required to show this field in inspection plan characteristics.

"Details" tab**"Gage" group****Gage**

Defines whether a gage or gage group is to be assigned to the characteristic:

Assignment of the gage to be used (or even gage group).

The resource management may be accessed using the resource type "PRM". The authorization "iriscp.gage" is required to show this field in inspection plan characteristics.

Gage designation

Shows the gage designation

"Properties" group

Certificate printing

The corresponding definition determines whether this characteristic is to be printed (show selection or print always) or not (print never) when certificates are printed out at a later stage (e.g. acceptance, inspection certificate). If the "display selection" option is chosen, a list of the characteristics that are assigned to this flag is displayed prior to printing. These characteristics have been predefined for printing in the list. However, this selection may be removed. Finally, all selected characteristics and the characteristics assigned to the "print always" option are printed out on the certificate. Characteristics assigned to the "print always" flag do not appear in a selected list, as they are printed in any case. However, it is important to note that these certificate flags only affect certificate forms.

Error weighting

For information purposes, a rating can be made here if the inspection result for this characteristic is "fail".

"Inspect" group

Analysis selection catalog

Using this selection function, an analysis selection catalog may be selected. This catalog restricts the selection options when failures are entered (failure types, failure locations, etc.). (But nevertheless, all available failures may be entered by directly entering their number).

Designation of analysis selection

Shows the designations of analysis selection catalogs

"Specifications" tab

Once the "specifications" tab has been selected, the sample scheme and dimensions may be entered. In this context, it has to be considered that (as already mentioned) the definition of tolerance limits in the master data of characteristics is only reasonable if certain conditions are met. The same applies to the definition or calculation of action and warning limits. The different possibilities are described anyway.

"Sampling scheme" group

Sampling scheme

The following sampling schemes are available:

- 100% inspection
- k value inspection
- Batch inspection
- n-c inspection
- SPC inspection

The sampling scheme determines the course of the inspection. With an n-c inspection with the parameters 5-0, 5 items are checked and 0 failures may be detected.

The section entitled "Sampling scheme" describes this process in more detail.

Sample size/assumable sample size

Specification of the sample size or assumable sample size depending on the sampling scheme, see section "sampling schemes".

Acceptance quantity

Acceptance quantity for the n-c inspection, please also see section "sampling schemes".

Interval type

Input for SPC or n-c inspections: time, pieces, once, none. Please also see the section "sampling schemes".

Interval value

Specifies the interval subject to the interval unit.

The authorization "iriscp.interval" is required to show this field in inspection plan characteristics.

Interval unit

For n-c or SPC inspections, e.g. minutes, hours.

The authorization "iriscp.interval" is required to show this field in inspection plan characteristics.

With output batch change

Causes an inspection to become due when output batches change.

The authorization "iriscp.interval" is required to show this field in inspection plan characteristics.

With machine status change

Causes an inspection to become due when machines change.

The authorization "iriscp.interval" is required to show this field in inspection plan characteristics.

Source status

Specifies the source statuses, separated by commas, (specific, non-productive machine statuses), which are to cause an inspection to become due when switching into a productive machine status.

The authorization "iriscp.interval" is required to show this field in inspection plan characteristics.

With the change of shifts

Causes an inspection to become due when shifts change.

The authorization "iriscp.interval" is required to show this field in inspection plan characteristics.

"Construction measures" group**Unit**

Pieces, meter, kg, etc. Unit of the characteristic. Units are assigned by accessing the unit catalog.

Decimal places

Number of decimal places. Leading zeros in front of the comma are not displayed in the specification fields. The number of decimal places that is defined within system settings is indicated by default.

Size

Plausibility and tolerance limits can be entered as absolute, relative or percentage values. Please note that relative or percentage lower limits (lower tolerance limit, lower process limits) have to be indicated with a negative algebraic sign.

Norm

Computation of tolerances on the basis of specific norms (e.g. ISO metric fits). Subject to the selected norm, further information is requested (e.g. engineering fit). Tolerance limits are computed automatically on the basis of these specifications.

Engineering fit

Computation of tolerance limits on the basis of a specific norm and fit. The selected fit depends on the selected norm.

UPL

Specifies the upper plausibility limit

UTL

Specifies the upper tolerance limit (upper specification limit)

Target value

Indicates the target value

LTL

Specifies the lower tolerance limit (lower specification limit)

LPL

Defines the lower plausibility limit

Generate error (UTL)/(LTL)

If the checkbox "generate error" is enabled it is defined whether a violated limit value automatically (in the background) results in the failure type "limit value violation" (AUTO:TG> or AUTO:TG<) to be defined when measured values are recorded. This option is not available for attributive characteristics, as the specification is only used for information purposes in this case.

"User fields" tab

In case user fields are defined for characteristics, they are displayed and may be edited here.

"Chart 1/chart 2" tab

The control charts to be used can be defined in the "chart 1/chart 2" tab. These control charts are later available in the measurement recording for terminals (AIP). Altogether two different control charts may be defined, whereas the action limits, warning limits and the mean value may be defined for each control chart if variable characteristics are in use. There are two different possibilities to define these limit values. They are either entered manually or determined automatically on the basis of specific default values that are defined in the "default values" tab "chart 1/2". For further information on control charts, please see the sections entitled "control chart for variable characteristics" and "control charts for attributive characteristics".

Chart 1 / chart 2

Defines the control chart that is to be displayed in the measurement recording dialog at the terminal. Action limits may also be defined on the basis of the control chart type.

Upper AL

Specifies the upper action limit. It can be determined by the system on the basis of default values, provided that the "calculate" checkbox is enabled. (Please also see section "control charts for variable characteristics").

Upper WL

Defines the upper warning limit. It can be determined by the system on the basis of default values, provided that the "calculate" checkbox is enabled. (Please also see section "control charts for variable characteristics").

MV (Mean value)

Specifies a mean value, e.g. as basis for the automatic calculation of limits by the system.

Lower WL

Defines the lower warning limit. It can be determined by the system on the basis of default values, provided that the "calculate" checkbox is enabled. (Please also see section "control charts for variable characteristics").

Lower AL

Specifies the lower action limit. It can be determined by the system on the basis of default values, provided that the "calculate" checkbox is enabled. (Please also see section "control charts for variable characteristics").

Generate trend error

The option "generate trend error" has to be activated to be able to generate an automatic error if a trend exists (e.g. seven values in a row are descending or ascending, the number of values is defined while the system is customized). To determine a trend, the samples of an inspection step characteristic are taken into account, sorted by the sample number, regardless on which machine data has been recorded.

Generate error (UWL) / (LWL)

If the checkbox "generate error (UWL) / (LWL)" is enabled it is defined whether a violated limit value automatically (in the background) results in the failure type "limit value violation" (AUTO:WG> or AUTO:WG<) to be defined when measured values are recorded. A violation of the limit value refers in this case to the defined control chart. If, for example, an xq chart is defined an automatic error will only be generated, provided that the corresponding xq value of the control chart, not the single value, is beyond the warning limits.

Generate error (UAL) / (LAL)

If the checkbox "generate error (UAL) / (LAL)" is enabled it is defined whether a violated limit value automatically (in the background) results in the failure type "limit value violation" (AUTO:EG> or AUTO:EG<) to be defined when measured values are recorded. A violation of the limit value refers in this case to the defined control chart. If, for example, an xq chart is defined an automatic error will only be generated, provided that the corresponding xq value of the control chart, not the single value, is beyond the action limits.

"Default values chart 1 / default values chart 2" tab

For further information on control charts, please see the sections entitled "control charts for variable characteristics" and "control charts" for attributive characteristics.

"Default for calculating limit values" tab**Calculation type**

Default values to compute limit values: Cpk, Sigma, sQuer/an, RQuer/dn, relative deviation of XQuer, percentage deviation of XQuer

Cpk

Default value of cpk

Sigma

Default value or computed sigma value

RQuer /sQuer

Default value for RQuer/sQuer

"Non-action probability" group**Action limits (non-action probability)**

Selects the action probability (only visible with the computation type: cpk, Sigma, RQuer /sQuer)

Warning limits (non-action probability)

Selects the action probability (only visible with the computation type: cpk, Sigma, RQuer /sQuer)

"Deviation from XQuer specification" group**rel. AL**

Direct entry of the action limits (only visible if the calculation types "rel. deviation/deviation in percent" are used)

rel. WL

Direct entry of the warning limits (only visible if the calculation types "rel. deviation/deviation in percent" are used).

"Confidence interval" group**Confidence interval**

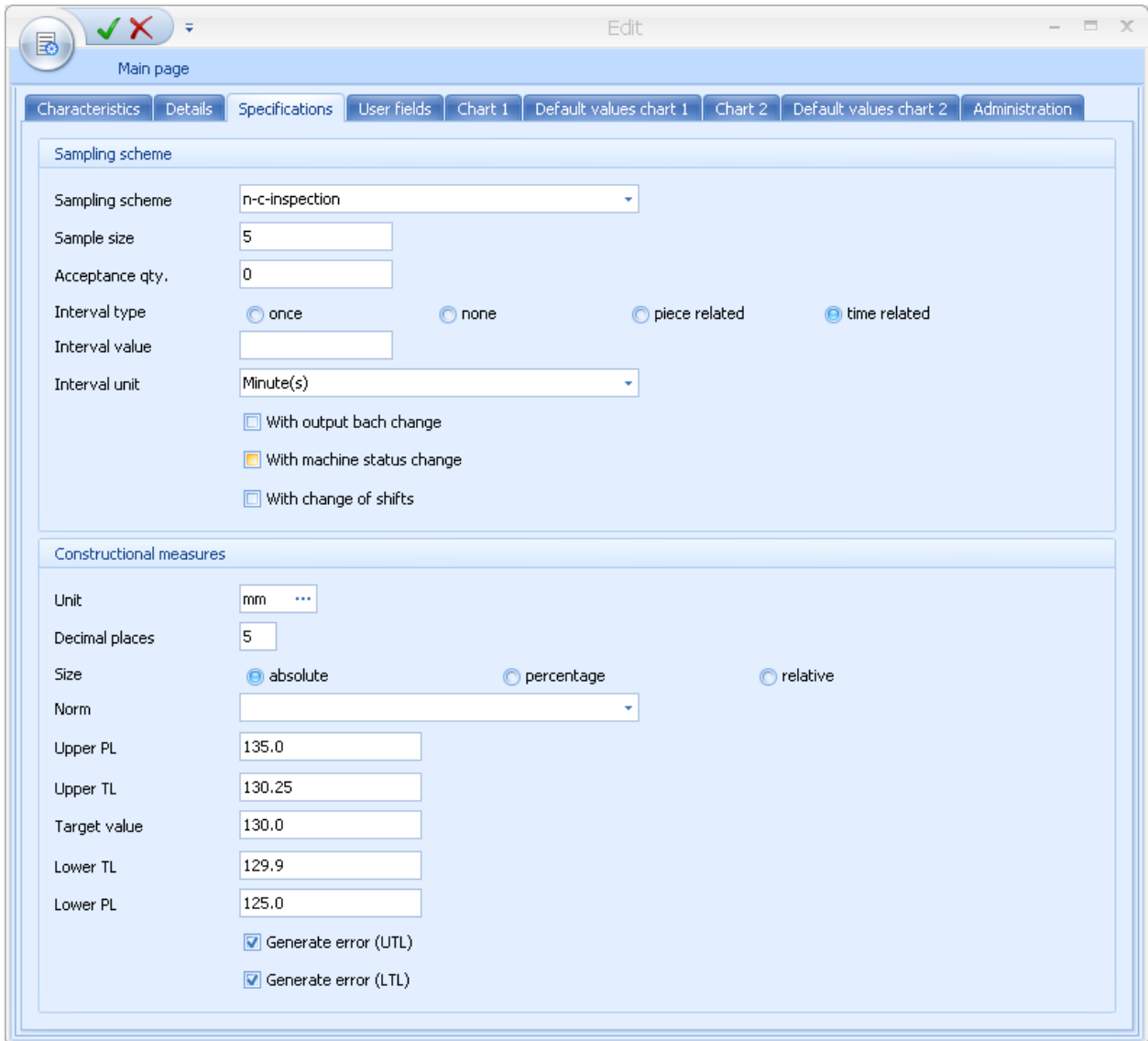
One-sided or two-sided. It may be chosen between "on-sided" and "two-sided" for the control charts R and s.

"XQuer" group**XQ**

Target value, tolerance center, mean value of xq chart, input (only visible and can only be selected with an xq control chart)

Editing functions

The below screenshot is one example for how an editing dialog could look like. The design and alignment of fields may deviate from that of this screenshot.



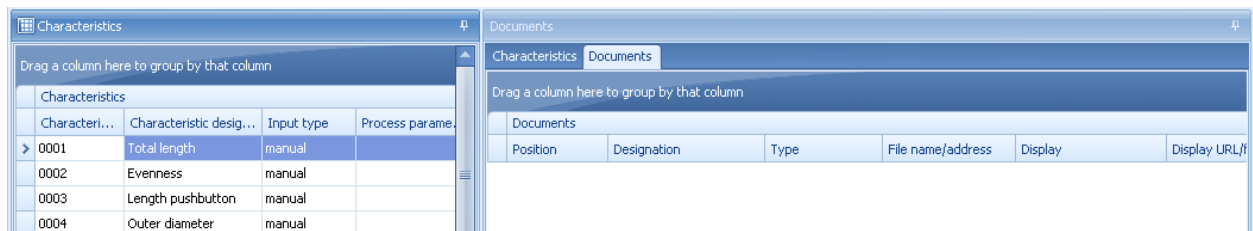
The screenshot shows an 'Edit' dialog box with the following fields and options:

- Sampling scheme:**
 - Sampling scheme: n-c-inspection
 - Sample size: 5
 - Acceptance qty.: 0
 - Interval type: ☐ once, ☐ none, ☐ piece related, ☒ time related
 - Interval value:
 - Interval unit: Minute(s)
 - ☐ With output bach change
 - ☐ With machine status change
 - ☐ With change of shifts
- Constructional measures:**
 - Unit: mm
 - Decimal places: 5
 - Size: ☒ absolute, ☐ percentage, ☐ relative
 - Norm:
 - Upper PL: 135.0
 - Upper TL: 130.25
 - Target value: 130.0
 - Lower TL: 129.9
 - Lower PL: 125.0
 - ☒ Generate error (UTL)
 - ☒ Generate error (LTL)

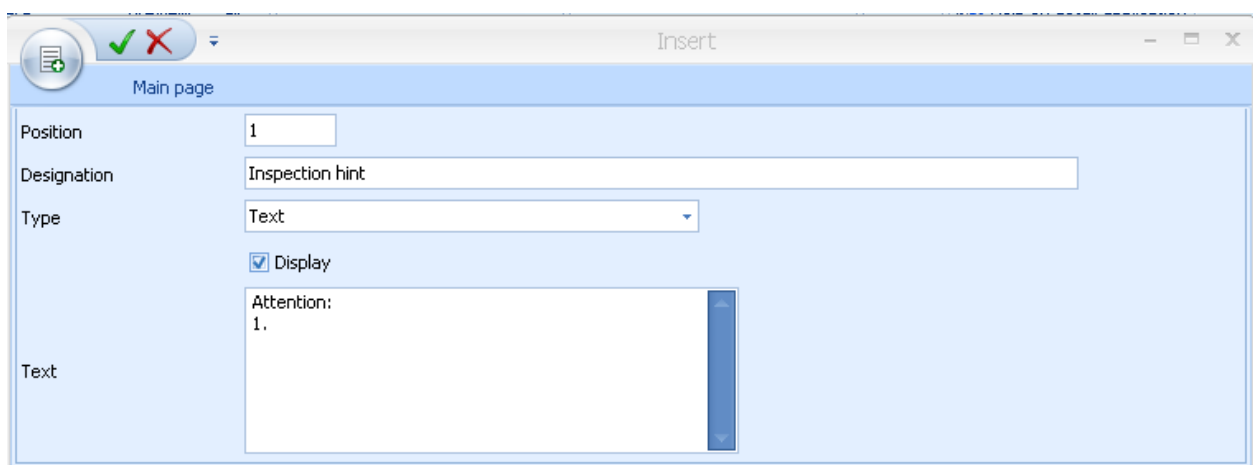
Toolbar

There are no further specific function keys in addition to the standard features.

"Documents" detail application



Provided that the "documents" tab has been activated, as many documents as required may be assigned to each characteristic. If this tab is activated corresponding editing buttons are enabled in the toolbar to edit the documents.



When documents are assigned, all formats registered by Windows are provided. Consequently, it is possible to assign simple documents (e.g. written in Word), drawings of any format and videos. However, the corresponding programs that are able to display the required formats have to be installed. In this context, the documents are opened by the program that has been linked in Windows.

"URL" and "text" are provided as file types. The file name including the path may be entered for the "file" type. The file type "URL" allows for the Internet or intranet to be accessed. The third file type "text" allows for the text to be entered directly.

Please note:

The different variants of the file type "URL" supported by the shop floor client are described in the relevant manual dealing with the shop floor client. It might, for example, be the case that "https" URL entries can be displayed on MOC but not on the AIP shop floor client.

A designation may be assigned to each defined document. Moreover, it can be determined in which order the documents are to be listed. This is made by the "position" field (numeric input). The specifications have to be unique within this list. In addition, the "display with inspection" checkbox determines whether or not the document can be displayed during the inspection process.

In this case as well it has to be reflected thoroughly whether documents are to be assigned without a precise reference to an article. Normally, document assignment depends on the article.

“Documents” toolbar

In addition to the standard features, the “Documents” detail application also provides a button to view the documents.



Show documents

With this button it is possible to open and show a document, which is attached by a link. In this case, however, a program needs to be installed that is able to open the linked file type, e.g. Acrobat Reader for PDF documents.

6.1 Sampling scheme

The user can choose from five sampling schemes in a specified list. Subject to the selected sampling scheme, some additional information has to be defined. Depending on how they are used in the different inspection plan areas (e.g. production, goods receipt, goods issue) only a subset of these sampling schemes is available.

Sampling scheme n-c inspection: The sample size is entered in the "sample size" field (= n) and the maximum number of admissible non-conforming units is entered in the "acceptance quantity" (=c) field. The figure "c" is defined as acceptance number. This means: if n = 50 und c = 1 the characteristic, and thus the item, is only rated as "fail" if two non-confirming units occur, provided that the sample size amounts to 50.

Sampling scheme 100% inspection: The sampling scheme "100% inspection" is generally only used in the goods receipt and goods issue. In this case, the sample size is computed from the actual quantity of the inspection requirement and corresponds to it.

Sampling scheme SPC inspection: The sampling scheme "SPC inspection" nearly corresponds to the "n-c" inspection plan. The only difference is that the acceptance limit "c" is not used in this case.

Sampling scheme batch inspection: The sampling scheme "batch inspection" only applies to the areas "goods receipt" and "goods issue" in the standard configuration. The percentage specifying how much percent of the batch is to be checked is entered here. Later in the inspection order characteristic the sample size is calculated from the actual quantity of the inspection requirement and multiplied by the specified percentage.

Provided that action limits are to be calculated, the expected (assumable) sample size has also to be entered here.

Sampling scheme k-value inspection: With the k value inspection the entered k value is checked against the computed k value and if this value is violated the sample is rated "fail".

6.2 Control charts for variable characteristics

The variable characteristics provide the charts xq, s and R.

The production dispersion is used in many computations of the statistical quality assurance. One example is the calculation of capability indices and action limits of a quality control chart. Vice versa, it is possible to estimate the dispersion of production, provided that a process capability index is indicated, and the action limits are calculated on this basis.

The specifications for the calculation of limit values can be found in the tab "default values chart 1" or "default values chart 2", where values to estimate the production dispersion can be entered. The action and warning limits can be calculated on the basis of these specifications. However, it is also possible to enter the production dispersion directly. This program provides altogether three computation options.

At first, the specifications are described on the basis of the xq and s chart. The differences with the R chart are explained in more detail in the sections that follow.

There is often a specification for the process capability index cpk. This specification is reasonable, as respecting the process capability index cpk implies that items can be produced within the range of tolerance. On the basis of the default cpk value, the system computes internally an estimated value for Sigma, which is entered to the right of the "Sigma" option for information purposes. This computed basic estimated value has in turn been designed to calculate the limit values of the xq/s chart. The computation is performed, once further data has been entered using the "calculate" button. The calculation method "cpk" is set by default. In addition, there are also the calculation methods "sigma" and sQuer/an".

This cpk value of 1,33 ensures that 99.725% of the characteristic values are within the tolerance. However, it also often required that 99.994% of the characteristic values are within the tolerance limit, which corresponds to a cpk value of 1.67.

The calculation method sq/a_n means that an estimate of the standard deviation is calculated from the quotient of the medium standard deviation and a correction factor a_n . This correction factor depends on the sample size, which is expressed by the index n . The values for a_n are defined in the system and are requested automatically. This estimate of the standard deviation is best in case that there is no specification of the process capability index and the production dispersion is unknown and thus the specification of the sq -value has still to be corrected by a correction factor. Moreover, it is the most efficient one under the given conditions. The sq -value has also to be specified for a subsequent calculation of limit values. This can be entered in the field on the right next to the sq/a_n option. If the “calculate” button is clicked in the preceding dialog the estimate sq/a_n is calculated from the specification of sq and it is entered on the right next to the sigma option for information. This estimate is then the basis for the calculation of action and warning limits of the xq/s chart

The third calculation method requires a sigma value being specified. In this case it is assumed that sigma is known and consequently the correction factor is not required. To the right of the “sigma” option the sigma value is entered. In contrast to the previous method sq/a_n is replaced by sigma. In cases of doubt and since in the majority of cases sigma is not really known, the calculation method based on the specification of sq including the automatic determination of the estimate sq/a_n should be performed for variances.

If the “relative deviation from Xq ” or the “deviation from Xq in percent” is selected as “specification for calculating limit values” the input option for “action probability in %” disappears and the “deviation from target value” can be entered instead. Limit values are then calculated on the basis of these values and the default calculation of Xq (target value, tolerance center, mean value of XQ chart, input).

Further details have to be made in order to determine action and warning limits of the Xq chart. An Xq value has to be specified. The system offers the possibilities to equate the Xq value with the tolerance center or the target value or to specify a value manually. If the process is supposed to be aligned to the mean value the tolerance center should be preferred as Xq value.

The action probability must be entered in per cent in order to calculate action and warning limits of the Xq/S -chart. For this purpose, it has to be defined beforehand whether the calculation is to be made on the basis of one-sided or two-sided limit values. One of the two options has to be selected.

Having defined the “one-sided” or “two-sided” option, the action probability has to be entered in per cent. All possible and reasonable action probabilities are defined in the system and need only to be selected from the list. The computation of the Xq chart is based on normal distribution. The value 99,725 is to be selected if, e.g., 99,725% of characteristic values are supposed to lie within the action limits. As the specification of a sigma area is commonly used by some users, the corresponding sigma area is displayed along with the respective action probability for information.

If warning limits are also to be calculated an action probability has also to be entered here. Please take into account that the probability value of the warning limit has to be less than the action limit value.

No sigma area is displayed in the selection list since the distribution of χ^2 is considered for the calculation of limit values of the s-chart. Apart from that, the entry is the same as for the Xq chart.

The calculation of limits is triggered as soon as the specifications made are saved, provided that the "calculate" checkbox has been clicked before.

As already mentioned, the user is also able to enter the limit values directly without any defaults.

If the R-chart is selected instead of the Xq or s-chart, the option sq/a_n is replaced by Rq/d_n in the specifications for the calculation. The calculation of the estimate of sigma is made on the basis of the default mean range R divided by the correction factor d_n . This correction factor in turn depends on the sample size n. The corresponding values are defined in the system and are selected automatically. Apart from that, the rest is the same as for the Xq or s-chart. Please note that limit values are not calculated on the basis of a χ^2 distribution for the R-chart but on the basis of a table defined in the system, which is based on standardized ranges.

6.3 Control charts for attributive characteristics

The p- and u-charts are available for attributive characteristics.

p designates the share of non-conforming units in the sample and u designates the failures per unit in the sample. It is important for the p-chart that each item is either described as error-free or faulty. If an item shows several failures it is referred to as faulty only once.

In contrast to variable characteristics there are no lower limit values. Furthermore, it is normally not necessary to state the values UTL, LTL and target value.

A pq or uq value has to be entered in per cent to calculate specifications automatically. This can be done in the default values dialog.

The calculation of limits is triggered as soon as the specifications made are saved, provided that the "calculate" checkbox has been clicked before.

Calculation is respectively based on normal distribution. The value 99,725 has to be selected if, e.g. 99,725% of the characteristic values are supposed to lie below the upper action limit. As the specification of a sigma area is commonly used by some users the corresponding sigma area is displayed with the respective action probability for information.

6.4 Computation of formulas

On the basis of an entered formula, measured values can be calculated automatically by way of measured values or statistic values of other characteristics that have been checked before.

The first part of the formula specifies where the calculation of formulas is supposed to take place. The following types are available:

- V – Calculation on the level of single values (Value).
A single value is generated for the calculated characteristic for each single value of the characteristics involved.
- S - Calculation on the level of samples (Sample).
Exactly one single value is generated for the calculated characteristic for each sample of the characteristics involved.
- C - Calculation on the level of characteristics (Criteria).
Exactly one single value is generated for the calculated characteristic (with respect to the overall statistic of all characteristics involved)

The actual formula follows this identifier. The following operators, functions and constants are supported in this context:

Functions	
abs(x)	Calculates the absolute value
atan(x)	Calculates the arc tangent
cosh(x)	Calculates the hyperbolic cosine
float(x)	Converts the value into a floating point number
sqrt(x)	Calculates the square root
acos(x)	Calculates the arc cosine
atan2(y,x)	Calculates the arc tangent of y/x
exp(x)	Calculates the exponential value
log(x)	Calculates natural logarithms
sin(x)	Calculates the sine
tan(x)	Calculates the tangent
asin(x)	Calculates the arc sine
cos(x)	Calculates the cosine
int(x)	Converts the value into an integer
log10(x)	Calculates common logarithms
round(x)	Rounds to integer value
round(x,y)	Rounds the value x to y decimal places
sinh(x)	Calculates the hyperbolic sine
tanh(x)	Calculates the hyperbolic tangent
trunc(x)	Reduces the value x to an integer value
trunc(x,y)	Reduces the value x to y decimal places
Operators	
x + y	Addition
x – y	Subtraction
x / y	Division
x * y	Multiplication
x ** y	Calculates x raised to the power of y
Absolute term (constant)	
pi	3.141592654
e	2.718281828

If constant, numeric values are used in formulas it has to be taken into account that no separators are used for thousands digits. If these constants are floating-point numbers it has to be considered that they use a dot as decimal separator instead of a comma.

The following syntax applies for the variables identifying the single values or statistic values of the order characteristics involved [x:y:z]. The x parameter identifies the statistic value to be used. The available values are listed as follows. Please consider possible restrictions regarding the calculation level.

- **X** – Single value
(is only available for calculations on the level of single values)
- **AVG** – Mean value
(is only available for calculations on the level of samples or characteristics)
- **MIN** – Minimum
(is only available for calculations on the level of samples or characteristics)
- **MAX** – Maximum
(is only available for calculations on the level of samples or characteristics)
- **SUMX** – Sum of single values
(is only available for calculations on the level of samples or characteristics)
- **R** – Range
(is only available for calculations on the level of samples or characteristics)
- **S** – Standard deviation
(is only available for calculations on the level of samples or characteristics)
- **N** – Sample size
(is only available for calculations on the level of samples or characteristics)
- **M** – Number of samples
(is only available for calculations on the level of characteristics)

The y parameter describes how the corresponding characteristic is supposed to be identified. The following possibilities are available:

- **SENO** – identification via the MS (maintenance sequence/operation sequence) of the characteristic (serial number)

- INCR – Identification via the characteristic number (inspection criteria)

If the characteristic number is not unique within the inspection requirement it is not predictable which one of the applicable characteristics is used at the time of calculation.

The parameter *z* identifies the characteristic on the basis of the field content determined by parameter *y*. Either the MS/OP number or the characteristic number of the calculation source is written in this field. If the characteristic number has a blank it should be replaced by an underscore within the formula.

Example 1:

A new characteristic is computed from the single values of the characteristic assigned to the number "LENGTH"/"LAENGE" divided by 2.5. A corresponding single value is supposed to be calculated for each single value of the source characteristic (calculation on the level of single values).

→ Formula: V: [X:INCR:LAENGE] / 2.5

Example 2:

The characteristic "surface" results from the product of the characteristics with the characteristic number "LENGTH"/"LAENGE" and "WIDTH TOTAL"/"BREITE GES". A single value of the characteristic "surface" is supposed to be calculated for each single value of both source characteristics (calculation on the level of single values).

→ Formula: V: [X:INCR:LAENGE] * [X:INCR:BREITE_GES]

Example 3:

The characteristic "maximum margin width" results from the subtraction of the minimum of the characteristic "inside diameter" (MS 10) from the maximum of the characteristic "outside diameter" (MS 20). A single value of the characteristic "maximum margin width" is supposed to be calculated for each sample of both source characteristics (calculation on the basis of samples).

→ Formula: S: [MAX:SENO:20] - [MIN:SENO:10]

The escalation CPAUMW.CALCULATED_CRITERIAS_GET_VARIABLE_VALUE is triggered if unknown variables are used within a formula (faulty parameters x and/or y). However, the escalation management module has to be licensed in this case.

When it comes to the calculation of formulas, it is allowed to compute new calculated characteristics from calculated characteristics that have already been computed. However, this nesting may not have more than 10 references one below the other. Furthermore, double concatenations must not be created (Example: characteristic A is calculated from characteristic B and characteristic C; characteristic C is calculated from characteristic A).

Tool numbers, machine numbers or cavity numbers, etc. are not defined for the single values computed.

The same number needs to be assigned to all source samples of the calculation to take over a corresponding number (batch number, sample number, serial number, etc.). If there is no number that is assigned to all source samples a number will not be assigned to the calculated sample. Provided that several numbers are found that are assigned to all source samples only the first number that has been found is assigned to the calculated sample. This function only applies to numbers, which have been assigned on sample level.

7 Specification List

1.1 Summary

Menu	Master data → Quality management → Specification list
Transaction code	sclq
Function authorization	sclq

Utilization

The specification list has been designed to create specifications, e.g. within the framework of family inspection planning.

To be able to use the specifications of the specification list for the generation of an inspection step with inspection step characteristics, the property "from list" has to be set in the corresponding inspection plan characteristics. In this case, only active entries of the specification list are used. A specification list entry is searched on the basis of the following key fields, when generating an inspection step.

- Area
- Article number, drawing issue number
- Resource number
- Characteristic no.
- Operation number and operation designation
- Customer no.
- Supplier number

The order of searching the specification list may be configured while customizing the system.

The specification list does not replace inspection planning. It rather is a supplement to family inspection plans.

Integration

This function is a fundamental components of family/group inspection planning, as the inspection specifications that vary with each item in the article group of a group inspection plan are defined in the specification list.

Prerequisite

An inspection plan for the article group including corresponding characteristics referring to the specification list has to exist to be able to use specification lists (configuration within inspection plan characteristics: "from list").

Selection criteria

The application provides the following selection criteria:

Area

List of available CAQ areas

Specification no.

Unique specification number

Version no.

Unique version within the specification. Only active, provided that the version control function has been activated within system settings.

Active

Active or inactive entries are filtered.

Special case

Special cases/normal cases or both are filtered.

Article number

Article number of the specification list entry; can be selected from the catalog for article master data.

Article designation

Article designation

Customer number

Customer number can be selected from the company catalog

Customer name

Customer name

Supplier number

Supplier number can be selected from the company catalog

Supplier designation

Supplier name

Operation

Operation number

Workplace

Workstation, e.g. machine

Note: The definition of machine related specification list entries isn't supported from the standard.
This field can be used for customer specific implementations.

Characteristic no.

Characteristics number

Characteristic designation

Characteristic designation

Characteristic type

The characteristic type is filtered: variable, attributive, inspection chart

If several selection criteria are used overlapping results are displayed in the specification list entries.

Field descriptions**"Characteristic key" tab*****"Characteristic" group*****Area**

Specifies the area for which the specification list entries are to apply.

Specification no.

Number of the specification

Version no.

Version number of the specification

Active

Checkbox. Shows whether the entry is active or inactive.

Workplace

Indicates the workplace.

Note:: The definition of machine related specification list entries isn't supported from the standard. This field can be used for customer specific implementation in order to define machine related inspections for the same article. This might be required if e.g. machines of different types are used.

Machine designation

Designation of the workplace/machine

Resource

A resource (e.g. tool) may be indicated.

Designation

Name of the resource

Characteristic no.

Unique number of the characteristic

Characteristic designation

Name of the selected characteristic number

Operation

Number of the operation

Operation designation

Designation of the operation

"Properties" group**Characteristic type**

Variable, attributive or inspection chart

Input type

Specifies whether data is collected manually or automatically.

Special case

Indicates whether this characteristic does no longer need to be checked after x "pass" inspections in a row.

Number of pass inspections in a row

The number of "pass" inspections in a row that is required for the characteristic to be no longer required to be checked.

No characteristic

Specifies whether this characteristic is not required to be checked for the combination of the indicated key fields.

"Article" group**Article number**

Article number for this specification list entry

Article designation

Designation of the selected article number

Drawing issue number

Drawing issue number of the selected article number

"Companies" group**Customer number:**

Customer no.

Customer name

Customer name of the selected customer number

Supplier number

Supplier no.

Supplier name

Supplier name of the selected supplier number



The key fields of the "characteristic key" tab can no longer be changed if a list entry is changed (edited).

"Specifications" tab

Go to

The fields included in the "specifications" tab correspond to those of the characteristic master data and are described in the documentation [MOC_CharacteristicsQm.](#)

"Chart 1/2" tab



Go to

The fields included in the "chart 1/2" tab correspond to those of the characteristic master data and are described in the documentation [MOC_CharacteristicsQm.](#)

Default values chart 1/2



Go to

The fields included in the "default values chart 1/2" tab correspond to those of the characteristic master data and are described in the documentation [MOC_CharacteristicsQm.](#)

Editing functions

The following dialog opens to edit a data record:

Toolbar

The below-mentioned additional functions are available besides the standard functions.



Activate

Function authorization: sclq.active

Activates a specification list entry. A previously released version is automatically deactivated.

**Deactivate**

Function authorization: sclq.release

Deactivates a specification list entry. The specification list entry is no longer used.

8 Customers

Summary

Menu	Master data → Quality management → Customers
Transaction code	cto
Function authorization	cto

The customer catalog has been designed to edit/keep customers. Provided that there is an interface to a higher-level system (e.g. ERP system), customers can be created automatically via interface. As soon as a new customer is created or changed in the ERP system, for example, the customer data record is automatically created or changed in the customer catalog including the defined information.

Utilization

The customer number uniquely identifies customers in all QM applications that refer to the customer catalog. The customer catalog is used as basis in particular for inspection requirements of production and and the goods issue as well as for the complaint management.

The “customer number” field is the key field, i.e. if a new customer is saved it is verified whether a customer with this key information exists already.

By distinguishing between active and inactive customers, it may be defined whether or not they are available in certain selection lists. Thus, no complaint can be created for an inactive customer, for example. However, inactive customers may be evaluated at any time. Moreover, inactive customers can be reactivated at any time.

Extensive address and contact data can be defined in addition to the customer number and the designation.

If a customer is designated as “party in charge” they will be included in the selection list for the parties in charge. Such selection lists are integrated in different detail applications. The list of the parties in charge is accessed mainly in the complaint management function and when measures are generated.

Integration

Customer data is a global catalog that is used in many QM applications. The below list shows the applications referring to the customer catalog.

- External people
- Departments

- Production inspection planning
- Production inspection requirements
- Complaint management
- Failure mode analysis

Prerequisite

There are no special requirements.

Selection criteria

The address fields 1, 2, and 3 are available in addition to the customer number and customer name.

The active or inactive customers can be restricted using the filter field "inactive".

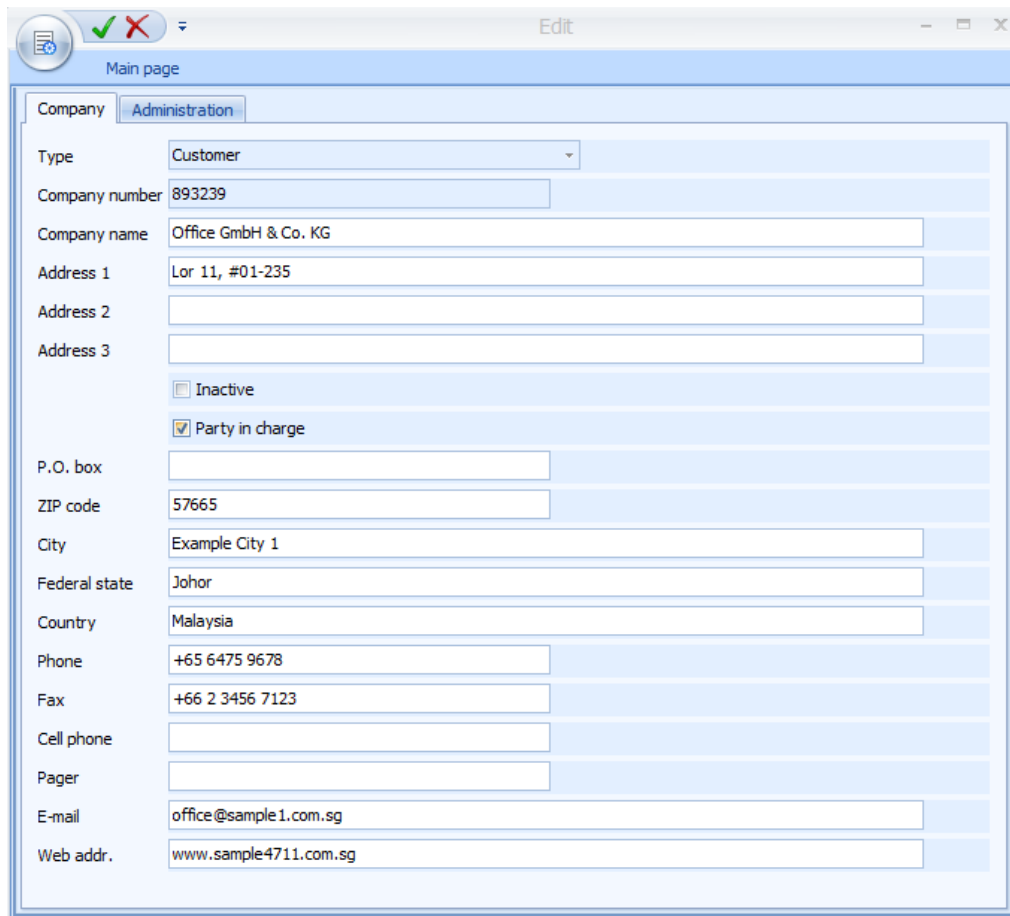
Field Descriptions

The available fields are self-explanatory and are not described separately, except for the address fields.

The content of the individual address fields is not specified by default and, as a result, may be defined by the user. Normally, address field 1 should include further details on the company, e.g. "site X". As there is no field defined for the street, address field 2 or 3 (to be preferred) is to be used for the street including street number.

Editing functions

The below dialog opens to edit a data record.



The screenshot shows a software window titled 'Edit' with a standard Windows-style title bar (minimize, maximize, close buttons). Inside the window, there is a 'Main page' tab. Below the tab, there are two sub-tabs: 'Company' and 'Administration', with 'Administration' currently selected. The form contains the following fields and controls:

- Type: A dropdown menu showing 'Customer'.
- Company number: A text field containing '893239'.
- Company name: A text field containing 'Office GmbH & Co. KG'.
- Address 1: A text field containing 'Lor 11, #01-235'.
- Address 2: An empty text field.
- Address 3: An empty text field.
- Inactive: A checkbox that is currently unchecked.
- Party in charge: A checkbox that is currently checked.
- P.O. box: An empty text field.
- ZIP code: A text field containing '57665'.
- City: A text field containing 'Example City 1'.
- Federal state: A text field containing 'Johor'.
- Country: A text field containing 'Malaysia'.
- Phone: A text field containing '+65 6475 9678'.
- Fax: A text field containing '+66 2 3456 7123'.
- Cell phone: An empty text field.
- Pager: An empty text field.
- E-mail: A text field containing 'office@sample1.com.sg'.
- Web addr.: A text field containing 'www.sample4711.com.sg'.

Toolbar

There are no other special function buttons in addition to the standard functions.

9 Manufacturer

Summary

Menu	Master data → Quality management → Manufacturer
Transaction code	mft
Function authorization	mft

The catalog of manufacturers has been designed to edit and update the list of manufacturers. Provided that there is an interface to a higher level system (e.g. ERP system) the manufacturers can be created automatically by interface. As soon as a new manufacturer is created or an existing manufacturer is changed in the ERP system, the corresponding manufacturer record is automatically created or changed with the defined information in the catalog of manufacturers.

Utilization

The manufacturer number uniquely identifies the manufacturers in all QM applications that refer to the catalog of manufacturers. In particular, the inspection requests of the goods receipt refer to the catalog of manufacturers.

The "manufacturer number" field is the key field, i.e. while saving a new manufacturer, the system checks whether there is already a manufacturer with this key information.

By differentiating between active and inactive manufacturers, it can be defined whether or not they are still to be available in certain selection lists. Consequently, it is, for example, impossible to create a goods receipt inspection request for an inactive manufacturer. However, it is possible to evaluate inactive manufacturers at any time. Moreover, inactive manufacturers may be reactivated at any time.

Extensive address and contact data may be defined in addition to the manufacturer number and designation.

If the manufacturer is designated as being a "party in charge" by clicking the corresponding field, the manufacturer is included in the selection list for the parties responsible. Such selection lists are integrated in different detail applications. The list of the parties responsible is mainly used within complaint management and for the creation of measures.

Integration

Manufacturer data is a global catalog that is used in many QM applications. Please find below some applications that refer to the manufacturer catalog.

- External people

- Departments
- Goods receipt inspection planning
- Goods receipt inspection requests
- Failure mode analysis

Prerequisite

There are no special requirements.

Selection criteria

The address fields 1, 2 and 3 are available in addition to the manufacturer number and the designation.

The "inactive" filter field allows for the data set to be restricted to active or inactive manufacturers.

Field descriptions

The available fields are self-explanatory and are not explained separately, except for the address fields.

The content of the individual address fields is not specified and, as a result, may be defined by the user. Normally, the address field 1 should include an addition to the company's name, e.g. "site X". As no field has explicitly been defined for the street, the address field 2 or 3 (to be preferred) is to be used for entering the street and street number.

Editing functions

The following dialog opens to edit a data record.

The screenshot shows a software window titled 'Edit' with a standard Windows-style title bar (minimize, maximize, close buttons). Inside the window, there's a 'Main page' tab. Below the tab, there are two sub-tabs: 'Company' and 'Administration', with 'Administration' currently selected. The form contains the following fields and controls:

- Type:** A dropdown menu set to 'Manufacturer'.
- Company number:** A text box containing '827363'.
- Company name:** A text box containing 'Spanner International Pte Ltd'.
- Address 1:** A text box containing '5 Ayer Rajah Crescent'.
- Address 2:** An empty text box.
- Address 3:** An empty text box.
- Inactive:** An unchecked checkbox.
- Party in charge:** A checked checkbox.
- P.O. box:** An empty text box.
- ZIP code:** A text box containing '62783'.
- City:** A text box containing 'Example City 2'.
- Federal state:** A text box containing 'Sarawak'.
- Country:** A text box containing 'Malaysia'.
- Phone:** A text box containing '+65 6341 8679'.
- Fax:** A text box containing '+66 2 6574 9900'.
- Cell phone:** An empty text box.
- Pager:** An empty text box.
- E-mail:** A text box containing 'info@sample45.com.sg'.
- Web addr.:** A text box containing 'www.sample4712.com.sg'.

Toolbar

There are no other special function buttons in addition to the standard functions.

10 Suppliers

Summary

Menu	Master data → Quality management → Suppliers
Transaction code	sup
Function authorization	sup

The supplier catalog has been designed to edit and update the list of suppliers. Provided that there is an interface to a higher-level system (e.g. ERP system), suppliers may be created automatically by the interface. As soon as a new supplier is created or an existing supplier is changed in the ERP system, the data record for this supplier is automatically created or changed including the specified information within the HYDRA-CAQ supplier catalog.

Utilization

The supplier number uniquely identifies suppliers in all QM applications that access the supplier catalog. The supplier catalog is used, in particular, for inspection requests of goods receipt and the complaint management.

The "supplier number" field is the key field, i.e. while saving a new supplier, the system checks whether or not there is already a supplier with this key information.

By distinguishing between active and inactive suppliers, it may be defined whether or not the suppliers are available in certain selection lists. Consequently, it is, for example, impossible to create a goods receipt inspection request for an inactive supplier. However, inactive suppliers may be evaluated at any time. Moreover, inactive suppliers may also be reactivated at any time.

In addition to the supplier number and designation, it is possible to specify comprehensive address and contact details.

Being identified as "party in charge", this supplier is included in the selection list for responsible parties. Such selection lists are integrated in different detail applications. The list of the parties responsible is mainly used within complaint management and for the creation of measures.

Integration

Supplier data is a global catalog that is used in many QM applications. Please find below some applications that refer to the supplier catalog.

- External people
- Departments

- Goods receipt inspection planning
- Goods receipt inspection requests
- Complaints management
- Failure mode analysis

Prerequisite

There are no special requirements.

Selection criteria

The address fields 1, 2 and 3 are available in addition to the supplier number and the designation.

The "inactive" filter field allows for the data set to be restricted to active or inactive suppliers.

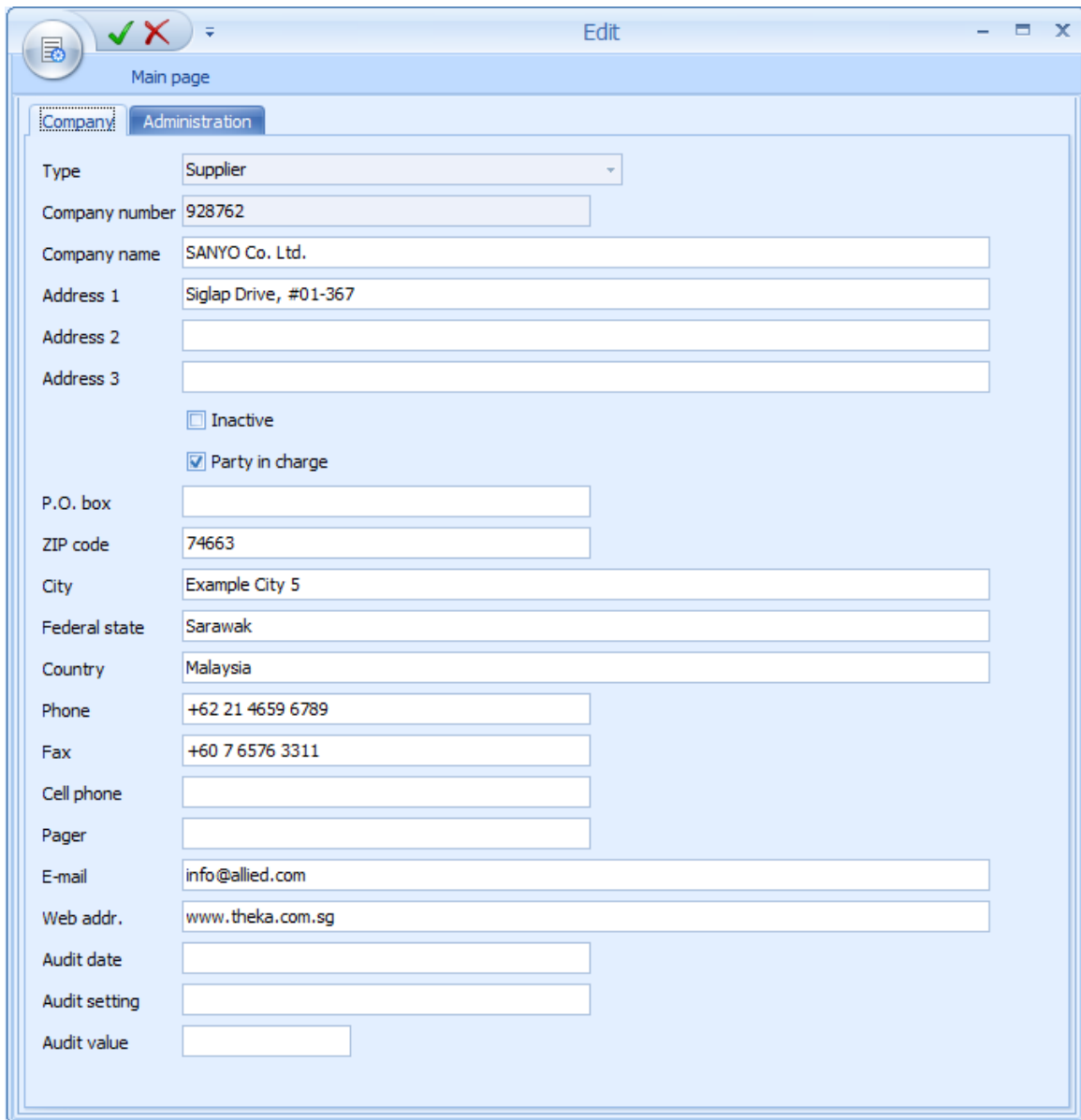
Field descriptions

The available fields are self-explanatory and are not explained separately, except for the address fields.

The content of the individual address fields is not specified explicitly and, as a result, may be defined by the user. Normally, the address field 1 should include an addition to the company's name, e.g. "site X". As no field has explicitly been defined for the street, the address field 2 or 3 (to be preferred) is to be used for entering the street and street number.

Editing functions

The following dialog opens to edit a data record.



The screenshot shows a software window titled "Edit" with a standard Windows-style title bar (minimize, maximize, close buttons). Inside the window, there is a "Main page" header. Below this, there are two tabs: "Company" (selected) and "Administration". The "Company" tab contains a form with the following fields and values:

Field	Value
Type	Supplier
Company number	928762
Company name	SANYO Co. Ltd.
Address 1	Siglap Drive, #01-367
Address 2	
Address 3	
Inactive	☐
Party in charge	☒
P.O. box	
ZIP code	74663
City	Example City 5
Federal state	Sarawak
Country	Malaysia
Phone	+62 21 4659 6789
Fax	+60 7 6576 3311
Cell phone	
Pager	
E-mail	info@allied.com
Web addr.	www.theka.com.sg
Audit date	
Audit setting	
Audit value	

Toolbar

There are no other special function buttons in addition to the standard functions.

11 Failure Types

Summary

Menu	Master data → Quality management → Failure types
Transaction code	ftyp
Function authorization	ftyp

The catalog of failure types has been designed to describe the occurred deviations in more detail, e.g. deviations from specified limit values. In addition to the other failure catalogs (failure location, failure cause, causer), the failure type catalog is very important as it also includes inactive failure types by default. These inactive failure types are required for the generation of automatic failure types, e.g. in case a limit value has not been respected. For this reason, these inactive failure types must not be deleted. All inactive failure types that are important for the automatic generation of failures start with the ID number "AUTO:". There are, for example, automatic failure types for

- Non-observance of the upper tolerance limit
- Non-observance of the lower tolerance limit
- Non-observance of action and warning limits

Failures the designation of which includes the number sign (#) automatically trigger the generation of a complaint if they are assigned to a measured value of one of the areas in-production inspection, goods receipt inspection or goods issue inspection.

Utilization

The "failure analysis number" field is the key field, i.e. when saving a new failure type, the system checks whether there is already a failure type with this key information.

The input of failure types is easy to handle. Only a failure analysis number and a corresponding designation have to be assigned.

Failure type groups may optionally be defined beforehand. Consequently, the corresponding group can be assigned to the respective failure type. This option should not be missed out as it provides improved reports/evaluations. Groups can be assigned by opening the group tree using the magnifier function. The hierarchical tree entries of the group tree allow for the requested group to be selected and taken over. Then the "groups" field of the editing dialog for failure types shows the assigned group including the hierarchical group structure.

If failure types are presented in list form the group hierarchy is represented by the columns "group 1" to "group 5".

Groups are edited in the "failure type groups" application, which is described in the manual entitled "[MOC_Groups.pdf](#)".

Under certain circumstances, it might be reasonable and recommendable to use a self-explanatory structure for the failure type number as failure key.

By differentiating between active and inactive failure types, it can be defined whether or not they are still to be available in failure selection lists in the later data acquisition process. However, it is also possible to evaluate inactive failure types at any time. Moreover, inactive failure types can be reactivated at any time.

Integration

The failure types catalog is used, among other things, within measurement recording and in complaint management. If deviations are detected in measurement recording the failure catalogs help describe the deviations in more detail and represent it in a way that allows for analyses/reports to be performed. Only in this way is it possible to determine failure mode analyses, take appropriate action (measures) and prevent the deviation from reoccurring.

In addition, this catalog is the basis for failure mode analyses relating to the failure types.

This catalog is also required for the creation of analysis selection catalogs as well as for using inspection chart characteristics within inspection planning and measurement recording. An analysis selection catalog includes a subset of all failures and restricts the list of failure (types) that can be selected during the collection process to those failures of the analysis selection catalog assigned to the characteristic. Consequently, the analysis selection catalog also determines the list of failure types for inspection chart characteristics.

Prerequisite

Functional requirements from other applications do not have to be met to be able to use this function.

Selection criteria

Selection criteria are self-explanatory and are not described separately. Failure types of a group can be filtered in the "groups" tab using the icon  and selecting a failure type group (in tree structure). The group tree list provides a function to cancel and accept the entries made.

The "inactive" filter field allows for the data set to be restricted to active or inactive failure types.

Field descriptions

The available fields are self-explanatory and are not explained separately.

The "inactive" check box identifies failure types that are no longer to be used in the active data acquisition process.

In a tree structure the group field shows the assigned group or allows for groups to be assigned in form of the tree structure.

Toolbar

There are no other special function buttons in addition to the standard functions.

12 Failure Locations

Summary

Menu	Master data → Quality management → Failure locations
Transaction code	floc
Function authorization	floc

The catalog of failure locations has been designed to describe the deviations occurred in more detail, e.g. deviations from specified limit values.

Utilization

The "failure analysis number" field is the key field, i.e. when saving a new failure location, the system checks whether there is already a failure location with this key information.

The input of failure locations is easy to handle. A failure analysis number and a corresponding designation only have to be assigned.

Failure location groups may optionally be defined beforehand. Consequently, the corresponding group can be assigned to the respective failure location. This option should not be missed out as it provides improved reports/evaluations. Groups can be assigned by opening the group tree using the magnifier function. The requested group may be selected and taken over in the group tree by way of the hierarchical tree entries. The assigned group including the hierarchical group structure then appears in the "groups" field of the editing dialog.

If failure locations are presented in list form the group hierarchy is represented by the columns "group 1" to "group 5".

Groups are edited in the "failure location groups" application, which is described in the manual entitled "[MOC_Groups.pdf](#)".

Under certain circumstances, it might be reasonable and recommendable to use a self-explanatory structure for the failure location number as failure key.

By differentiating between active and inactive failure locations, it can be defined whether or not they are still to be available in failure selection lists in the later data acquisition process. However, it is also possible to evaluate inactive failure locations at any time. Moreover, inactive failure locations can be reactivated at any time.

Integration

The failure location catalog is used, among other things, within measurement recording and in complaint management. If deviations are detected in measurement recording the failure catalogs help describe the deviation in more detail and represent it in a way that allows for analyses/reports to be performed. Only in this way is it possible to determine failure mode analyses, take appropriate action (measures) and prevent the deviation from reoccurring.

In addition this catalog is the basis for failure mode analyses relating to the failure locations.

The failure locations of this catalog can also be integrated in analysis selection catalogs. An analysis selection catalog includes a subset of all failures and restricts the list of failure (locations) that can be selected during the collection process to those failures of the analysis selection catalog that is assigned to the characteristic.

Prerequisite

Functional requirements from other applications do not have to be met to be able to use this function.

Selection criteria

Selection criteria are self-explanatory and are not described separately. Failure locations of a group can be filtered in the "groups" tab clicking the icon  and selecting a failure location group (in tree structure). The group tree list provides a function to cancel and accept the entries made.

The "inactive" filter field allows for the data set to be restricted to active or inactive failure locations.

Field descriptions

The available fields are self-explanatory and are not explained separately.

The "inactive" check box identifies failure locations that are no longer to be used in the active data acquisition process.

In a tree structure the group field shows the assigned group or allows for groups to be assigned in form of a tree structure.

Toolbar

There are no other special function buttons in addition to the standard functions.

13 Failure Causes

Summary

Menu	Master data → Quality management → Failure causes
Transaction code	fcau
Function authorization	fcau

The catalog of failure causes has been designed to describe the occurred defects/failures in more detail, e.g. if limit values have been infringed.

Utilization

The "failure analysis number" field is the key field, i.e. when saving a new failure cause, the system checks whether there is already a failure cause with this key information.

The input of failure causes is easy to handle. A failure analysis number and a corresponding designation only have to be assigned.

Failure cause groups may optionally be defined beforehand. Consequently, the corresponding group can be assigned to the respective failure cause. This option should not be missed out as it provides improved reports/evaluations. Groups can be assigned by opening the group tree using the magnifier function. The requested group may be selected and taken over in the group tree by way of the hierarchical tree entries. The assigned group including the hierarchical group structure then appears in the "groups" field of the editing dialog.

If failure causes are presented in list form the group hierarchy is represented by the columns "group 1" to "group 5".

Groups are edited in the "failure causes groups" application, which is described in the manual entitled "[MOC Groups.pdf](#)".

Under certain circumstances it might be reasonable and recommendable to use a self-explanatory structure for the failure cause number as failure key.

By differentiating between active and inactive failure causes, it can be defined whether or not they are still to be available in failure selection lists in the later data acquisition process. However, it is also possible to evaluate inactive failure causes at any time. Moreover, inactive failure causes can be reactivated at any time.

Integration

The failure cause catalog is used, among other things, within measurement recording and in complaint management. If deviations with respect to tolerance limits are detected in measurement recording the failure cause catalog helps describe the actual failure cause in more detail and represents it in a way that allows for analyses/reports to be made. Only in this way is it possible to determine failure mode analyses, take appropriate action (measures) and prevent the deviation from reoccurring.

In addition this catalog is the basis for failure mode analyses relating to the failure cause.

The failure causes of this catalog can also be integrated in analysis selection catalogs. An analysis selection catalog includes a subset of all failures and restricts the list of failure (causes) that can be selected during the collection process to those of the analysis selection catalog assigned to the characteristic.

Prerequisite

Functional requirements from other applications do not have to be met to be able to use this function.

Selection criteria

Selection criteria are self-explanatory and are not described separately. Failure causes of a group can be filtered in the "groups" tab using the  icon and selecting a failure cause group (in tree structure). The group tree list provides a function to cancel and accept the entries made.

The "inactive" filter field allows for the data set to be restricted to active or inactive failure causes.

Field descriptions

The available fields are self-explanatory and are not explained separately.

The "inactive" check box identifies failure causes that are no longer to be used in the active data acquisition process.

In a tree structure the group field shows the assigned group or allows for groups to be assigned in form of a tree structure.

Toolbar

There are no other special function buttons in addition to the standard functions.

14 Causers

Summary

Menu	Master data → Quality management → Originator
Transaction code	ori
Function authorization	ori

The catalog of causers/originators has been designed to describe the cause for an initial situation. In this context, the causer or originator does not imperatively have to be a person or a group of people. A machine can also be the causer/originator.

Utilization

The "causer number" field is the key field, i.e. while saving a new causer/originator, the system checks whether there is already a causer with this key information.

The input of causers is easy to handle. Only an ID number and a corresponding designation have to be assigned.

Causer groups may optionally be defined in advance and the corresponding group can be assigned to the originator. This option should not be missed out as it provides improved reports/evaluations. Groups can be assigned by opening the group tree using the magnifier function. The requested group may be selected and taken over in the group tree by way of the hierarchical tree entries. The assigned group including the hierarchical group structure can then be found in the "groups" field of the editing dialog for causers.

If causers are presented in list form the group hierarchy is represented by the columns "group 1" to "group 5".

Groups are edited in the "causer groups" application, which is described in the manual entitled [MOC Groups.pdf](#).

Under certain circumstances, it might be reasonable and recommendable to use a self-explanatory structure for the originator number as analysis key.

By distinguishing between active and inactive causers, it can be defined whether or not they are still to be available in selection lists for the causers in the later data acquisition process. However, it is possible to evaluate inactive originators at any time. Moreover, inactive causers can be reactivated at any time.

Integration

The catalog of originators is used, among other things, within measurement recording and in complaint management. A causer/originator may be added if defects are detected and documented within measurement recording. Only in this way is it possible to determine the basic causers, take appropriate action (measures) and prevent the defects from recurring during the failure mode analysis that follows.

The causers/originators of this catalog can also be integrated in analysis selection catalogs. An analysis selection catalog includes a subset of all failures and causers/originators and restricts the list of failures and causers that can be selected during the collection process to those of the analysis selection catalog assigned to the characteristic.

Prerequisite

Functional requirements from other applications do not have to be met to be able to use this function.

Selection criteria

Selection criteria are self-explanatory and are not described separately. Causers/originators of a group can be filtered in the "groups" tab using the icon  and selecting a causer group (in tree structure). The group tree list provides a function to cancel and accept the entries made.

The "inactive" filter field allows for the data set to be restricted to active or inactive causers.

Field descriptions

The available fields are self-explanatory and are not explained separately.

The "inactive" check box identifies causers that are no longer to be used in the active data acquisition process.

In a tree structure the group field shows the assigned group or allows for groups to be assigned in form of the tree structure.

Toolbar

There are no other special function buttons in addition to the standard functions.

15 Measures

Summary

Menu	Master data → Quality management → Measures
Transaction code	meas
Function authorization	meas

The catalog of measures has been designed to define the action to be taken, for example, to prevent the detected failures from recurring or to allow for them to be identified in time.

Utilization

The "measure number" field is the key field, i.e. while saving a new measure, the system checks whether there is already a measure with this key information.

The input of measures is easy to handle. Only a measure number and a corresponding designation have to be assigned.

Measure groups may optionally be defined beforehand and the corresponding group can be assigned to the measure. This option should not be missed out as it provides improved reports/evaluations or structures. Groups can be assigned by opening the group tree using the magnifier function. The requested group may be selected and taken over in the group tree by way of the hierarchical tree entries. The assigned group including the hierarchical group structure can then be found in the "groups" field of the editing dialog for the measures.

If measures are presented in list form the group hierarchy is represented by the columns "group 1" to "group 5".

Groups are edited in the "measure groups" application, which is described in the manual entitled [MOC_Groups.pdf](#).

By distinguishing between active and inactive measures, it can be defined whether or not they are still to be available in selection lists for the measures in the later data acquisition process. Inactive measures can be reactivated at any time.

Integration

The measures catalog is used, among other things, within measurement recording and in complaint management. If deviations are detected within measurement recording actions can be triggered immediately on the basis of the catalog of measures. At first the assigned measures have the status "open". In complaint management a party responsible (e.g. a person) as well as a deadline may be defined. In addition to this, an actual date, the completion in % and effectiveness in % may be specified as well. Normally, these fields are only filled out, once a measure has been completed (measure status switches to "done").

Subject to whether the corresponding license for the escalation management module has been purchased, the measures can also be sent to the defined person by e-mail.

Prerequisite

Functional requirements from other applications do not have to be met to be able to use this function.

Selection criteria

Selection criteria are self-explanatory and are not described separately. Measures of a group can be filtered in the "groups" tab using the icon  and selecting a measure group (in tree structure). The group tree list provides a function to cancel and accept the entries made.

The "inactive" filter field allows for the data set to be restricted to active or inactive measures.

Field descriptions

The available fields are self-explanatory and are not explained separately.

The "inactive" check box identifies measures that are no longer to be used in the active data acquisition process.

In a tree structure the group field shows the assigned group or allows for groups to be assigned in form of the tree structure.

Toolbar

There are no other special function buttons in addition to the standard functions.

16 Analysis Selection

Summary

Menu	Master data → Quality management → Analysis selection
Transaction code	asc
Function authorization	asc

Analysis selection catalogs allow for the set of failure types, failure locations, failure causes, originators, and measures that are available in measurement recording to be restricted specifically.

The overall application is divided into two hierarchical levels. A master-detail grid is used for presentation. The data records of the analysis selection are created on the first level. One level below, the corresponding failure types, locations, causes, originators and measures are assigned to these data records of the analysis selection. There is a separate tab for each assignment type on this second level.

Separate function keys to create, edit or delete data records are available on each level of the master-detail grid.

Utilization

The analysis selection number identifies the data records of the analysis selection uniquely in all QM applications in which they may be selected and assigned.

A data record of an analysis selection catalog only consists of a number and designation as well as of the flag to disable it ("inactive" field). The most crucial factor here is the possibility to assign failure types, locations, causes, originators and measures to a data record of the analysis selection.

If a characteristic is assigned an analysis selection catalog only the failure types, failure locations, failure causes, originators and measures listed in this catalog will be available for this characteristic when measured values are recorded. Consequently, the failure list, for example, can be designed in relation to characteristics.

Analysis selection catalogs are mainly used for the assignment to an inspection chart characteristic. An assignment is almost mandatory for inspection chart characteristics, as in any other case, the user may choose from the whole set of failure types of the entire master data catalog when recording measured values for this characteristic. This would make inspections confusing and too complex.

Integration

Analysis selection catalogs are used in all applications dealing with characteristics. By assigning analysis selection catalogs in these applications, the selection list for failures, originators and measures is restricted for measurement recording.

Prerequisite

The master data for failure types, failure locations, failure causes, originators and measures need to be maintained before this function can be used in a useful manner.

Selection criteria

Selection criteria are self-explanatory and are not described separately.

Field descriptions

The available fields are self-explanatory and are not explained separately.

The check box "inactive" identifies data records of the analysis selection that are no longer to be used for the definition of characteristics (of inspection plans/inspection orders).

17 Inspection Planning

Summary

Menu	Quality management → In-production inspection → Inspection planning Quality management → Goods receipt → Inspection planning Quality management → Goods issue → Inspection planning Quality management → Initial sample → Inspection planning Quality management → Gage management → Inspection planning
Transaction code	iplp
Function authorization	iplp

Utilization

Subject to the field of application, inspection planning is the most important function, as inspection plans are the basis for the generation of inspection orders and the resulting inspections. This means inspections in the area of goods receipt, production and goods issue as well as initial sample inspections and the calibration of test equipment and measurement equipment. As the requirements as regards inspection planning are nearly identical in all areas, the HYDRA inspection planning functions are also nearly identical for these areas.

In creating inspection plans, the user is supported in most cases by the master data catalogs. For this reason it is very important to conscientiously maintain this master data.

Integration

Inspection planning is a prerequisite to generating inspection requirements, the corresponding inspection steps and, as a result, the relevant inspections/calibrations.

The inspection plan in use is referenced in the generated inspection requirements. Moreover, some evaluations/reports also refer to inspection planning. Consequently, data of the same inspection plan number can be added to the basis of a control chart for inspection order characteristics. The same applies to the global control chart analysis, where special inspection plan filter fields may be used. The global control chart analysis does not take into account calibration data and, as a result, no filtered calibration inspection plans.

Another essential relation has been established to the production control plan. As no production control plan can be created without having generated inspection plans beforehand.

Prerequisite

Relevant master data needs to be edited/maintained to be able to create inspection plans. Which master data has to be maintained depends on the respective field of application. However, it is a fundamental prerequisite that articles/items and characteristics are maintained beforehand. Article groups also need to be created, provided that inspection plans based on article groups are created.

Selection criteria

The paragraph that follows shows some of the available selection criteria. Self-explanatory filter options are not listed.

Area

Selection list of the configured areas of in-production inspection, goods receipt and goods issue. By default, the following areas are available.

In-production inspection: Production

Goods receipt: Goods receipt

Goods issue: Goods issue

Initial sample: Initial sample inspection

Gage management: Gage management

Active

By checking the corresponding field, the inspection plan list can be restricted to active inspection plans. If this checkbox is not checked the list only shows inspection plans that are in the "in process" and "released" status. The third state of this checkbox (grayed out) shows all inspection plans. This is set by default.

Article number

Filters the article number with respect to inspection plans based on articles.

Article designation

Filters the article name with respect to inspection plans based on articles.

Article group

The article group tree can be opened by clicking the  icon, in case inspection plans for article groups are to be filtered. A function to cancel and accept the entries made is provided.

Operation

Operation number

Customer number

Direct input or opening of the customer catalog and taking over of the number from the entry selected there.

Supplier number

Direct input or opening of the supplier catalog and taking over of the number from the entry selected there.

Manufacturer number

Direct input or opening of the manufacturer catalog and taking over of the number from the entry selected there.

Field descriptions***Inspection plan header*****Area, inspection plan number, inspection plan index**

Uniqueness of all existing inspection plans is secured by the "area", "inspection plan number" and "inspection plan index". The area may be selected. The inspection plan number and inspection plan index may be entered using alphanumeric characters. All three fields are mandatory fields.

The input of these three pieces of information must be unique, i.e. no other inspection plan may exist that already includes this information. By assigning a structured inspection plan number, it is possible to provide specific information. This information might be useful later during sorting. As an alternative, the inspection plan number may also be used synonymously with the article number or gage group number. The inspection plan index corresponds to the inspection plan version. If an existing inspection plan version is to be modified, yet it cannot simply be edited because it has already been used to generate inspection orders, it is recommended to copy the original inspection plan and then modify the inspection plan index (e.g. incrementing it by 1). The inspection plan number should be maintained, if possible, to facilitate later evaluations/reports (e.g. for displaying control charts applying to several orders, in which case the inspection plan versions on which the evaluation is based differ while the inspection plan number is identical).

Inspection plan type

The inspection plan types "article inspection plan" and "group inspection plan" are distinguished. The "article inspection plan" type allows for an article to be assigned from the article master data for which this inspection plan is to be created. The "group inspection plan" option has to be selected to create an inspection plan for article groups. This allows for an article group to be selected. The selected article group is displayed in a separate field within the tree structure.

Article

Input of the article number. Provided that it is known, it may be entered directly. Otherwise, the article catalog can be opened and the requested article can be identified and taken over using the filter and sort criteria provided there. By selecting an article, the drawing issue number, article designation, customer article number and the drawing number are taken over from the master data record and displayed in the corresponding fields.



The customer article number is not displayed within the goods receipt inspection planning.



Nothing may be entered here if the field "article number" or "article group" is shown in gage management.

Drawing issue number

The drawing issue number can be entered directly, just as it is the case for the article number.



In case the article number and the drawing issue number are entered directly, the master data record is determined on the basis of this information while saving and the article designation, customer article number as well as the drawing number are displayed.



The article is uniquely identified by the combination of article and drawing issue number. It is not obligatory to define and use a drawing issue number.



Nothing may be entered here if the "drawing issue number" field is shown in gage management.

Operation assignment

One of the two assignment types "an inspection plan for each OP" and "an inspection plan for all OPs" may be selected.

By assigning operations, it is defined whether the inspection plan to be created should include the characteristics of different operations ("one inspection plan for all OPs") or whether the inspection plan should only include the characteristics of one operation. The fields "operation" and "operation designation" are shown if the "an inspection plan for each OP" option is selected. In any other case, operation details are assigned in the area of inspection plan characteristics. The option "one inspection plan for all OPs" should be selected as far as possible. This allows, for example, printing out of all inspection characteristics of an article, although it has several operations.



The option "one inspection plan for all OP" should be used in areas that are not assigned to a production order. This provides the advantage that the "fictitious" operation only has to be entered once in the inspection plan header, instead for every inspection plan characteristic.

However, the system has to be configured accordingly by MPDV consultants to be able to use this inspection plan variant.

Operation, operation designation

If operations are used, only one of the two fields "operation" or "operation designation" needs to be filled out. Only the operation number has to be entered in the "operation" field if an inspection requirement/inspection order is to be generated automatically by logging an operation of a production order on.

It has to be taken into consideration that these fields are used as search criteria when inspection orders are generated at a later point in time. Provided that only the designation is entered for the operation within the inspection plan, exclusively this information needs to be provided when inspection orders are generated.



A "fictitious" operation has to be assigned (e.g. 9999) even in areas that are not assigned to a production order (e.g. in goods receipt or in gage management). This is necessary to generate inspection steps at a later point in time.

Released/active

Shows whether the inspection plan is "released" and/or "active". If the inspection plan is released or active the corresponding checkboxes are checked. An inspection plan is released and activated, i.e. its status is changed, only by using the corresponding toolbar functions. An inspection plan has to be released before it can be activated.



Only active inspection plans are taken into consideration, when inspection orders are generated at a later point in time.

Released by / on

Shows the HYDRA user who has released it. The release date is displayed additionally.

Valid from / until

Instead of an "unrestricted" activation (using the toolbar), a validity period may be entered here. This period is then taken into account when the inspection order is generated. Yet activation for a certain period means that the user has no clear overview of currently valid inspection plans, and it is therefore recommended to use the "global/unrestricted" activation option using the toolbar if possible. If activated via the toolbar functions, the system carefully monitors whether an active inspection plan exists for the specified article and checks whether this plan includes the same drawing issue number, customers and suppliers. If this is indeed the case, the previously active inspection plan is automatically deactivated.

Type of inspection

Inspection based on characteristics or pieces

When it comes to an inspection based on characteristics and, say a sample size of 5, the characteristic is checked completely first. Whereas with the inspection based on pieces, the characteristic is changed every time a measured value is collected, as items are checked entirely.

Action

Create inspection step or create and immediately release inspection step

Only inspection steps that have been released can be checked.

Customer / supplier / manufacturer

If a customer / manufacturer or supplier is selected this inspection plan only applies for the selected company. Consequently, the customer / manufacturer / supplier becomes the key field of an active inspection plan. In this case, the inspection requirement must also include a company entry in order for this inspection plan to be used for the generation of the inspection requirement.

Dynamic modification type

Characteristic-related, batch-related or none

The authorization "iriscp.dynamic" is required to show these fields.

Batch-related: In this case, it is defined on the level of inspection plan headers which transitional definition is to be used for the dynamic modification. In addition, the initial inspection severity is also defined on the level of inspection plan headers. Now only the reference to the dynamic modification norm is required on the level of inspection plan characteristics. In this context, however, the selection is restricted to those dynamic modification norms referring to the same inspection severity definition as the assigned transitional definition.

Characteristic-related: In this case, the transitional definition, initial inspection severity as well as the dynamic modification norm are referenced on the level of inspection plan characteristics. In this context, however, the inspection severity definition of the assigned transitional definition has to match the assigned dynamic modification norm.

None: The "dynamic modification" function has not been enabled for this inspection plan.

Transitional definition

The authorization "iriscp.dynamic" is required to show these fields.

This field is only available if a dynamic modification relating to batches or characteristics is selected. It provides the possible inspection severities and controls switching between the inspection severities.

Initial inspection severity

The authorization "iriscp.dynamic" is required to show these fields.

This field is only available if a dynamic modification relating to batches or characteristics is selected. It defines which inspection severity is used for checking the first goods received according to the basics for the dynamic modification history.

Initial sample form

The different form types available for initial sampling can be chosen here. The form type "VDA volume 2, 4th edition" is only supported at the moment. Subject to the selected form type, a corresponding list of categories is shown in the inspection plan characteristics.

Editing functions

The key fields "area", "inspection plan" and "inspection plan index" cannot be changed in the editing mode.

Toolbar



Copy

The below dialog opens to copy an inspection plan:

The target area type and target area may be entered here. Normally, the user should select an area that is identical to that of the source inspection plan. The new inspection plan number and inspection plan index are to be entered afterwards. In case a new version is generated from an existing inspection plan, normally the same inspection plan number is used and only the inspection plan index is changed.



Activate

Function authorization: `iplp.activate`

Makes the inspection plan status "active".



Deactivate

Function authorization: `iplp.deactive`

Puts an inspection plan that is in the "active" status back to the "released" status.



Release

Function authorization: `iplp.release`

Puts an inspection plan that is in the "in process" status to the "released" status.



In process

Function authorization: `iplp.inprocess`

Puts an inspection plan that is in the "released" status in the "in process" status.

"Print" detail application

Function authorization	<code>iplp.print</code>
------------------------	-------------------------

The print dialog of the inspection plan header opens a list of available reports. These are Word forms. The potential content of these forms is determined by the Web services that are available in the respective context. The form entries, i.e. the content of the form list of the corresponding print dialog, are defined within the master data of quality management. This is where the basis for new forms is established and the corresponding form properties are defined. A corresponding license is required to be able to change the forms with respect to content and design.

Toolbar - print

There are no other special function buttons in addition to the standard functions.

"Inspection plan characteristics" detail application

The detail application of inspection plan characteristics is nearly identical to the master data of characteristics application. For this reason, reference is made only to additional features in the following.



Go to

For further information on the definition of characteristics, please refer to the functional description of the document entitled [MOC_CharacteristicsQM](#).

On the level of inspection plan characteristics the corresponding inspection plan characteristics are assigned to the previously defined inspection plan header. Assignments are performed by way of initial data creation using the characteristics catalog and taking over the characteristic selected there. By accepting the characteristic, all master data entries are copied to the inspection plan characteristic. Every (copied) detail may be changed and / or complemented afterwards. The characteristic designation is often complemented to define it in more detail.

It is also possible to create a characteristic not included in the characteristics catalog. However, this is recommended only in exceptional cases, since all analyses (e.g. failure mode analysis) are based on characteristics found in the catalog. It is therefore preferable to properly maintain the characteristics catalog.

Different properties and settings can be defined, before specific characteristic data is complemented.

Field descriptions

Inspection plan characteristics

OP sequence

The operation sequence number (OP sequence/AFO) determines the order in which the inspection is later performed. Entries must be unique. In the ideal case, operation sequence numbering should be assigned in steps of 10. This provides the option of later inserting a new characteristic between two existing ones.

Characteristics number

Number of the characteristic selected from master data

Characteristic type

Decides whether the inspection is based on the collection of measured values (variable) or on the specification of the number of detected failures (attributive). When it comes to the attributive inspection, the decision is often only based on the "pass" or "fail" results. Further characteristic types are the inspection chart and the information characteristic. The information characteristic has only been designed to display a document while the inspection is running. Subject to the input type, the lower area of the dialog provides corresponding sampling schemes.

Inspection result base

Decides whether all samples or only the last sample is used to determine the inspection result (pass/fail).

Mandatory inspection

If this checkbox is activated, at least one measured value has to be entered for this characteristic, before an inspection order can be completed with this characteristic.

Calculate characteristic

Identifies a characteristic to be calculated

Formula

Further details can be taken from the manual dealing with the CAQ master data characteristics.

Initial sample creation

Shortlist of categories for initial sampling. The contents depend on the form type selected in the inspection plan header. Within the resulting initial sample report, the inspection plan characteristics are summarized by category and a separate page is printed for each group.

Copy characteristic

It may be defined here which characteristic is to be copied from an initial sample inspection plan to a production plan, for example. Only those characteristics are copied that have been marked relevant in this field.

Details

Defines whether - with respect to this characteristic - the contents of the "details" tab are defined within the inspection plan or whether they have to be determined from the available master data characteristic or an entry from the specification list when inspection requirements are later generated on the basis of this inspection plan. If the "from characteristic catalog" option is selected the "details" tab is hidden. The "from characteristics catalog" option is reasonable if it is a characteristic the specifications of which are to be defined identically for several articles (even applicable for several article groups). In this case, specifications can be defined centrally within the master data. This reduces the input and update efforts considerably.

Specifications

Defines whether - with respect to this characteristic - the contents of the "specification" tab are defined within the inspection plan or whether they have to be determined from the available master data characteristic or from a specification list entry, when inspection requirements are later generated on the basis of this inspection plan. The tabs "specifications", "chart 1", "default values chart 1", "chart 2" and "default values chart 2" are hidden if the option "from list" or "from characteristic catalog" is checked. The "from list" option is only reasonable if inspection plans for article groups are in use. The specification list is used for an inspection plan for article groups if the specifications vary from item to item for certain characteristics.

Dynamically modified

The authorization "iriscp.dynamic" is required to show these fields.

This field is only available if a dynamic modification relating to batches or characteristics has been activated in the inspection plan header. This option allows for definitions to be made that a characteristic is not to be modified dynamically, although the dynamic modification option is selected in the inspection plan header.

Transitional definition

The authorization "iriscp.dynamic" is required to show these fields.

This field is only available if a dynamic modification relating to batches or characteristics has been activated in the inspection plan header. The transitional definition that can be selected provides the possible inspection severities and controls switching between these different inspection severities. For the dynamic modification based on batches, the transitional definition is assigned in the inspection plan header and is not available on the level of inspection plan characteristics in this case.

Initial inspection severity

The authorization "iriscp.dynamic" is required to show these fields.

It defines which inspection severity is used for checking the first goods received according to the basics for the dynamic modification history. Only the inspection severities of the previously selected transitional definition are available. For the dynamic modification based on batches, the transitional definition is assigned in the inspection plan header and is not available on the level of inspection plan characteristics in this case.

Norm

The authorization "iriscp.dynamic" is required to show these fields.

This field is only available if a dynamic modification relating to batches or characteristics is activated in the inspection plan header and if this characteristic is actually characterized by being relevant to dynamic modification. The dynamic modification norm to be used is assigned from master data.

Inspection level

The authorization "iriscp.dynamic" is required to show these fields.

This field is only available if a dynamic modification relating to batches or characteristics is activated in the inspection plan header and if this characteristic is actually characterized by being relevant to dynamic modification. An inspection level can only be selected if the assigned norm corresponds to DIN ISO 3951 or DIN ISO 2859.

AQL

The authorization "iriscp.dynamic" is required to show these fields.

This field is only available if a dynamic modification relating to batches or characteristics is activated in the inspection plan header and if this characteristic is actually characterized by being relevant to dynamic modification. An AQL value can only be selected if the assigned norm corresponds to DIN ISO 3951 or DIN ISO 2859.

Method

The authorization "iriscp.dynamic" is required to show these fields.

This field is only available if a dynamic modification relating to batches or characteristics is activated in the inspection plan header and if this characteristic is actually characterized by being relevant to dynamic modification. A method can only be selected if the assigned norm corresponds to DIN ISO 3951.

**Go to**

The fields of the respective tabs correspond to those of the characteristics master data. Further information on these fields can be gathered from the functional description of the document entitled [MOC_CharacteristicsQM](#).

Inspection plan characteristics - toolbar**Copy**

Data of the selected characteristic is opened in insert mode to be able to copy inspection plan characteristics. All fields can be changed. An operation sequence number that does not yet exist in this inspection plan has to be assigned to be able to save it.

Main page

Characteristic

OP: 10

Characteristic no.: 0001

Characteristic designation: Total length

Machine:

Group:

Planned on: ☒ Machine ☐ Group

Characteristic type: ☐ attributive ☐ inspection hint ☒ variable

Inspection result base: ☐ all samples ☐ last sample

☐ Mandatory inspection

☒ Formula

Formula:

Inspection station: 03-001

Operation: 30

Operation designation: Assembly

"Inspection plan documents" and "documents of inspection plan characteristics" detail applications

Inspection plan

Drag a column here to group by that column

Inspection plan	Insp. plan index	Article	Drawing issue nu...	Released	Active	Status	Article designation
77 9938 882 01	77 9938 882	C		☒	☐	released	Metal Pen, 4-colour, 10000m
77 9938 882 02	77 9938 882	C		☒	☐	released	Metal Pen, 4-colour, 10000m

Insp. plan characteristics

Document

Position	Designation	Type	File name/address
1	reference model	file	metal_pen.jpg

Insp. plan document

Position: 1

Designation: reference model

Type:

☒ Display with inspection

The above screenshot shows how an inspection plan document is assigned

Inspection plan

Drag a column here to group by that column

Inspection plan	Insp. plan index	Article	Drawing issue nu...	Released	Active	Status	Article designation
77 9938 882 01	77 9938 882	C		☒	☐	released	Metal Pen, 4-colour, 10000m
77 9938 882 02	77 9938 882	C		☒	☐	released	Metal Pen, 4-colour, 10000m

Insp. plan characteristics

Characteristic

OP	Characteristic	Characteristic designation	Characteristic type	Inspection result base	Mandatory
10	0001	Total length	variable		

Characteristic documents

Position	Designation	Type	File name/address	Display wit...	Created by	Created
1	test example	file	TotalLength1.jpg	☒	Miller	27/12/20

Characteristic document

Position: 1

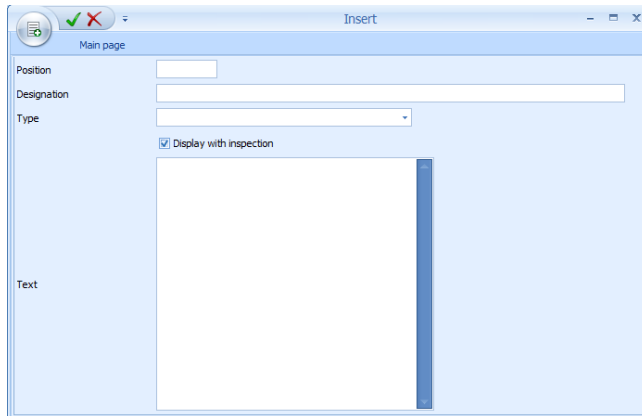
Designation: test example

Type:

☒ Display with inspection

The above screenshot shows how a document of an inspection plan characteristic is assigned

Provided that the "inspection plan documents" tab or the "characteristic documents" tab has been activated in the master detail grid, as many documents as required may be assigned to each inspection plan and each inspection plan characteristic. By enabling these tabs, the toolbar offers corresponding buttons to edit the documents.



When documents are assigned, all formats registered by Windows are provided. Consequently, it is possible to assign simple documents (e.g. written in Word), drawings of any format and videos. However, the corresponding programs that are able to display the required formats have to be installed. In this context, the documents are opened by the program that has been linked in Windows.

The file types are: "FILE", "URL" and "Text". The file name including path may be entered manually with the "file" type. The "URL" file type enables access to the Internet or Intranet. The third file type "Text" allows for text to be entered directly.

A designation may be assigned to each defined document. Moreover, it may be determined in which order the documents are to be listed. The "position" field is used for this purpose (numeric input). The specifications made within this list must be unique. In addition, the checkbox "display with inspection" determines whether or not the document may be shown during the inspection process.

"Inspection plan documents" and "documents of inspection plan characteristics" - toolbar

In addition to the standard features, there is also a button to display the documents.



Show documents

If a document link is defined this button opens and shows this document. However, a program, which is able to visualize the linked file type, has to be installed on the PC.

18 Inspection Requirements

Summary

Menu	Quality management → In-production inspection → Inspection requirement Quality management → Goods receipt → Inspection requirement Quality management → Goods issue → Inspection requirement Quality management → Initial sample → Inspection requirement Quality management → Gage management → Inspection requirement
Transaction code	irp ¹
Function authorization	irp

Utilization

The inspection requirement is as important as the production order is in the shop floor data collection module and includes one or several inspection steps (e.g. one inspection step for each operation). In this context, inspection steps result from the inspection plan search triggered by the generation of inspection requirements. For this reason, the inspection requirement rather includes general information, such as the order or article number. The inspection step that is subordinate to the inspection requirement finally includes the characteristics to be checked. Consequently, the inspection step or the operation of the production order (and, as a result, the inspection step as well) is selected and logged on to the terminal for measurement recording.

As the functional requirements are nearly identical in all areas (e.g. goods receipt, production, goods issue, gage management), the corresponding applications are structured nearly in the same way for these areas.



¹New inspection requirements cannot be created if the "inspection requirement" application is opened by using transaction codes.

In creating inspection requirements, the user is generally supported by master data catalogs. For this reason, it is very important to conscientiously maintain master data.

Integration

Inspection requests/inspection steps are generated on the basis of inspection plans that have been created beforehand and refer to them. When being generated, the contents of inspection plans are copied into the inspection request/inspection steps. Thus, the relation is only based on referencing. The limited modification options that are enabled in the inspection plan do not affect already generated inspection requirements/inspection steps.

The inspection steps, in turn, are the "objects" that can be checked/inspected and the operation number connects them with the corresponding operation of the production order within production. Within gage management, inspection steps can be put on a level with the actual calibration order.

Nearly all reports/evaluations refer to the data of inspection requests/inspection steps and the available inspection step characteristics. Restricting filters are, e.g. the article number, order number, operation, and inspection step characteristic.



In gage management, inspection requirements are generated automatically from the activity calendar by a user action. In exceptional cases, inspection requirements can also be generated directly in the "inspection requirement" application. In this case however, the inspection plan number has to be entered manually.

Prerequisite

Inspection plans need to be created beforehand in the same area/context to be able to create inspection requests/inspection steps. Master data is also required. Which master data has to be maintained depends on the respective field of application.

Selection criteria

The list that follows shows some of the available selection criteria. Self-explanatory filter options are not listed.

Area

Selection list of the configured areas of in-production inspection, goods receipt or goods issue. By default, the following areas are available:

In-production inspection: Production

Goods receipt: Goods receipt

Goods issue: Goods issue

Initial sample: Initial sample inspection

Gage management: Gage management

Inspection request number

The inspection requirement number uniquely identifies the inspection request. HYDRA automatically generates the inspection requirement number, which is unique.

Status

By opening a status list, the different inspection request statuses may be filtered. In this way, it is possible to restrict the list of inspection requirements, e.g. to the completed inspection requests.

Skip lot (available for the goods receipt and goods issue only)

Due to dynamic modifications, an inspection requirement can get the classification "skip lot" and, as result, it will not be checked.

Order

Number of the corresponding production order or calibration order.

Article number

Direct input or opening of the article catalog and taking over of the number from the entry selected there.

Drawing issue number

The article is identified uniquely by the combination of "article number" and "drawing issue number".

Customer number

Direct input or opening of the customer catalog and taking over of the number from the entry selected there.

Supplier number

Direct input or opening of the supplier catalog and taking over of the number from the entry selected there.

Manufacturer number

Direct input or opening of the manufacturer catalog and taking over of the number from the entry selected there.

Actual date from / until

The time may be restricted here. Normally, in production this field is not filled out. In the goods receipt area the "actual date" is the date when the goods are delivered.

Planned date from / until

The time may be restricted here. Normally, in production this field is not filled out. In the goods receipt sector the "planned date" defines the date when the goods are (initially) planned to be delivered.

Manufacturing date from / until

The time may be restricted here.

"Inspection request" field descriptions

The elementary fields are described below.

Inspection requirement**Area**

Area in which the inspection request is to be created. A selection list of the configured areas for in-production inspection, goods receipt, goods issue, initial sampling or gage management facilitates the creation process. By default, the following areas are available.

In-production inspection: Production

Goods receipt: Goods receipt

Goods issue: Goods issue

Initial sample: Initial sample inspection

Gage management: Gage management

This field is a key field for searching for the active inspection plan or the specified inspection plan including index on the basis of which the inspection request, inspection steps and inspection step characteristics are to be generated.

Inspection request number

The inspection request number is automatically generated upon saving. It is numeric, identifies an inspection request uniquely and cannot be changed.

Operation, operation designation

If operations are in use only one of the two fields "operation" or "operation designation" has to be filled out. Only the "operation number" field is to be filled out, provided that an inspection requirement and/or an inspection order is to be generated automatically by logging an operation of a production order on. The work plan structures cause the operation number to be referenced almost exclusively.

Subject to system configuration, this field can be a key field for searching for the active inspection plan on the basis of which the inspection requirement, inspection steps and inspection step characteristics are to be generated. If the system is configured accordingly, all inspection steps relating to the operations included in the relevant/detected inspection plan are generated already when the inspection requirement is generated.



Even areas that are not assigned to a production order (e.g. goods receipt and gage management), might require a "fictitious" operation (e.g. 9999) to be specified, subject to system configuration. This "fictitious" operation has to be included in the relevant/detected inspection plan.

Inspection plan number / inspection plan index (specified)

As an alternative to indicating key fields for searching for the active inspection plan to generate inspection requirements, the inspection plan that is to be used for the generation may also be entered in these fields.

Date

When a new data record is created, this field includes the system date, which may be changed afterwards. Once it has been saved, this field may no longer be changed.

Article

Input of the article number. Provided that it is known, it may be entered directly. Otherwise, the article catalog can be opened and the requested article can be identified and taken over using the filter and sort criteria provided there. By selecting an article, the drawing issue number, article designation, customer article number and the drawing number are taken over from the master data record and displayed in the corresponding fields.

This field is a key field for searching for the active inspection plan on the basis of which the inspection requirement, inspection steps and inspection step characteristics are to be generated, provided that no inspection plan including index has been specified.



This field is empty or should be left empty in gage management.

Drawing issue number

Like the article number, the drawing issue number can be entered directly.

This field is a key field for searching for the active inspection plan on the basis of which the inspection requirement, inspection steps and inspection step characteristics are to be generated, provided that no inspection plan including index has been specified.



In case the article number and the drawing issue number are entered directly, the master data record is determined on the basis of this information while saving and the article designation, customer article number as well as the drawing number are displayed.



The article is identified uniquely by the combination of article and drawing issue number. It is not obligatory to define and use a drawing issue number.



Provided that the “drawing issue number” field is shown in gage management, it is not allowed to assign any values to it here.

Companies

Customer / supplier / manufacturer

The corresponding company number may either directly be entered or by opening the corresponding master data catalog and selecting the corresponding entry. Once the inspection requirement has been selected and/or saved, the company name and address data are displayed additionally.

These fields are key fields for searching for the active inspection plan on the basis of which the inspection request, inspection steps and inspection step characteristics are to be generated, provided that no inspection plan including index has been specified.

Date - quantities

Actual quantity

The inspection scope is calculated subject to the selected sampling scheme and the actual quantity. This is especially true in goods receipt and goods issue if dynamic modification is enabled. Normally, in production this field remains empty.

Target quantity

In particular for the goods receipt, it is important to indicate the target quantity in order to determine the difference between the ordered quantity and the actually delivered quantity.

Scrap

This field is only relevant if HYDRA exclusively uses the CAQ functions and the shop floor data collection (BDE) module is not activated.

Rework

This field is only relevant if HYDRA exclusively uses the CAQ functions and the shop floor data collection (BDE) module is not activated.

Actually produced quantity (act. prod. qty.)

This field is only relevant if HYDRA exclusively uses the CAQ functions and the shop floor data collection (BDE) module is not activated.

Actual date

In particular for the goods receipt, it is important to indicate the actual date in order to determine the difference between the planned delivery date and the actual delivery date. Normally, in production this field remains empty.

Planned date

In particular for the goods receipt, it is important to indicate the planned date in order to determine the difference between the planned delivery date and the actual delivery date. Normally, in production this field remains empty.

Status - details**Status**

The status describes the inspection requirement status, e.g. whether it is completed, unchecked ("created" status) or whether inspection results already exist ("partial result" status).

Overall result

The overall result defines whether the inspection request is classified as "pass" or "fail".

Usage decision

The usage decision classifies the overall result in more detail. The list of usage decisions can be complemented when customizing the system. The usage decision is preset subject to the overall result when inspection requirements are completed. A "pass" result leads to the usage decision "release" and a "fail" result leads to the usage decision "reject". The preset usage decision may be changed when completing the inspection requirement manually.

Skip lot

Due to dynamic modifications, an inspection requirement can be classified as "skip lot" and, as a result, it will not be checked.

PPS/ERP reference

If inspection requirements are automatically generated by higher-level systems using an interface, this reference number allows for unique assignments to be made for uploading the results.

PPS/ERP addition

If inspection requirements are automatically generated by higher-level systems using an interface, another piece of information may be added to the unique reference number for uploading results. This field is only filled out by interface.

Uploaded to PPS/ERP

If this field is enabled (by interface) it indicates that the upload to the higher-level system has been performed and the higher-level system has confirmed this via interface. This field is only populated by interface.

Inspection plan number/inspection plan index (used)

The inspection plan number and inspection plan index make clear which inspection plan was used as basis for the generation of the inspection requirement and subordinate data (inspection steps, inspection step characteristics).

Cavity assignment

The authorization "ipl_cav" is required for displaying these fields.

The options "none" or "sample" are available for cavity assignment. These options derive from the inspection plan.

Dynamic modification**Dynamic modification type**

Shows the dynamic modification type of the respective inspection requirement. .

Batch-related: The available inspection plan has a dynamic modification based on batches.

Characteristic-related: The existing inspection plan has a dynamic modification based on characteristics. In this case, the transitional definition, the initial inspection severity as well as the dynamic modification norm are kept on the level of inspection step characteristics.

None: The available inspection plan does not include dynamic modification.

Transitional definition

These fields (number and designation) are only available if dynamic modification based on batches or characteristics is defined in the existing inspection plan. The transitional definition provides the possible inspection severities and controls switching between these different inspection severities. The content cannot be changed.

Determined inspection severity

It defines which inspection severity is/was used for checking the goods received according to the basics for the dynamic modification history. This field is only available if dynamic modification based on batches is defined in the existing inspection plan. The content cannot be changed.

Revised inspection severity

It defines which inspection severity is/was used for checking the goods received according to revising the inspection severity manually. This field is only available if dynamic modification based on batches is defined in the existing inspection plan.

Form**Party in charge type**

Provides the shortlist of the different categories of responsible parties, e.g. department, customer, supplier, external persons

Party in charge

List of the responsible parties restricted to the type selected beforehand. This list includes all the master data that were marked accordingly in the "party in charge" field.

Party in charge name

Only shows the name/designation of the selected responsible party.

Form

Form type of the available inspection plan.

Calibration**Resource type**

Shows the resource type of the resource to be calibrated. For gages, this is the type "PRM".

Resource

Shows the resource number (gage number) that is assigned to this inspection requirement.

Maintenance

Shows the activity number that is assigned automatically by creating an activity in the activity calendar. The relevant activity is determined or referenced based on this unique ID number.

Calibration findings

Shows the result entered or determined by completing the calibration process, e.g. capable without restrictions

Calibration status

Shows the status entered or determined by completing the calibration process, e.g. released

Inspection requirement - editing functions

The key fields "area" and "inspection requirement number" as well as the fields "skip lot" and "uploaded to ERP/PPS" cannot be changed in the editing mode.

Inspection requirement toolbar



Release

Function authorization: irp.release

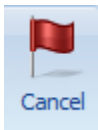
Transfers the inspection requirement to the status that existed at the time when it was completed, e.g. to the "partial result" status. This is required in cases where the inspection request has already been completed and is to be brought again into a condition that can be checked.



Complete

Function authorization: irp.complete

Opens a dialog to assign a usage decision, whereas a usage decision is already preset on the basis of the existing inspection results. The inspection request is finally transferred to the "completed" status, once it has been completed. The inspection steps pertaining to it can no longer be checked in this status.



Cancel

Function authorization: irp.cancel

Transfers the inspection requirement to the "canceled" status. This status no longer allows for the inspection steps pertaining to it to be checked.

"Print" detail application

Function authorization	irp.print
------------------------	-----------

The print dialog of the inspection requirement header opens a list of available reports. These are Word forms. The potential content of these forms is determined by the Web services that are available in the respective context. The form entries, i.e. the content of the list of forms of the corresponding print dialog, are defined within the master data of quality management. This is also where the basis for new forms is established and the corresponding form properties are defined. A corresponding license is required to be able to change the forms with respect to content and design.

The report, 4th edition, 2nd volume VDA (German Association of the Automotive Industry), is available as special form for initial sampling. In addition to the cover sheet, the relevant annex sheets of the inspection characteristic results are printed as well. Each appendix includes the inspection characteristics including the relevant results for a category, e.g. data for dimensional checks (category: dimension).

Print - toolbar

There are no other special function buttons in addition to the standard functions/features.

"Complete inspection requirement" detail application

Function authorization	irp.complete
------------------------	--------------

The corresponding button has to be clicked in the toolbar to complete an inspection request. An editing dialog opens afterwards.

Information about the inspection steps pertaining to the inspection requirement is displayed in addition to information about the inspection requirement itself (i.e. order, batch, article, company data). An inspection requirement cannot be completed as long as any inspection steps are still open, i.e. the field "without result" includes a value greater than zero. Within this dialog box it is, however, possible to complete all inspection orders that are still open by using the button "complete all inspection orders", provided that the conditions of a mandatory inspection that might exist are met. The result and further use of the inspection requirement may then be specified. If all of the inspection steps have been completed with "pass", the system proposes "pass" as the result and "release" as the possible use. If any inspection step has been completed with "fail", the system proposes "fail" as the result and "reject" as the possible use. Another result and use may still be selected though.

The list indicating the available usage decisions may be changed through MPDV customizing services.

"Inspection step" detail application

Proceeding from an inspection requirement in the master detail grid to an inspection requirement in the sub-level of "inspection steps" results in all inspection steps pertaining to the selected inspection requirement to be displayed. The detailed view of the selected inspection step additionally shows the inspection step data. The number or the combination of the inspection steps depends on the configurations made for the existing inspection plan. In case an inspection plan includes a configuration saying that the operation is defined on the level of inspection plan characteristics and, in addition to this, system configuration provides for the creation of all corresponding inspection steps along with the generation of the inspection requirement and the first inspection step, the inspection requirement includes an inspection step for every operation included in the inspection plan.

Moreover, it is possible to release, complete or cancel an inspection step using the toolbar functions.

Inspection steps may be deleted but not created or edited manually.

The current status of the inspection step can be taken from the "rating" category of the "inspection step" tab. The following statuses are possible.

Status	Description
No characteristic (KMM)	The inspection step has no characteristics able to be inspected. The inspection step is faulty and no inspection can be performed.
Created (ERS)	The inspection step has been created but not yet released for inspection. No measurement values have been recorded yet.
Inspection step released (FRE)	The inspection step has been released for inspection. No measurement values have been recorded yet.
Partial result (TER)	The inspection step has been released; some measurement values have been recorded.
Inspection in progress (IPR)	Measurement values for this inspection step are currently being recorded at another terminal.
Completed (ABG)	The inspection step has been completed.
Canceled (STO)	The inspection step has been canceled.
Skip lot (SKL)	The inspection step is in the "skip-lot" state.

"Inspection step" field descriptions

The essential inspection point fields are described in the paragraphs that follow. Self-explanatory fields are not explained.

Inspection step

Inspection step number

Unique number of the inspection step

Status

Status description of the inspection step, e.g. "completed", "partial result", etc.

Result

Qualitative classification of the inspection result based on the corresponding inspection results of the characteristics pertaining to it.

Workplace

Number of the workplace for which this inspection step has been generated.

Workplace designation

Name of the workplace for which this inspection step has been generated.

Inspection point identification

Up to nine different fields may be activated, which can or must be filled out later when inspection points are generated. By customizing the CAQ system options, it may be defined which fields are to be enabled, how they are labeled (for some fields) and whether or not it is a mandatory field.

If a field is activated and it is a mandatory field the corresponding checkbox will be "checked". The corresponding checkbox is "checked" and grayed out if it is not a mandatory field but enabled for the resulting data collection. If it is disabled, the checkbox of the field pertaining to it is not activated at all.

Once activated within the inspection point, the fields 1 to 3 are available for entering values with respect to "equipment", "functional location" and "physical sample". The fields 4 to 9 are user fields relating to inspection points. The fields 4 and 5 are "alphanumeric" fields, the fields 6 and 7 are numeric, the field 8 is a "date" field and field 9 is a "time" field.

Order data for the inspection step**Order**

This is e.g. the production order number in production.

Operation

Number of the operation for which this inspection step has been generated. The operation number taken from the existing inspection plan is the basis in this context.

Operation designation

Operation designation pertaining to the operation number taken from the existing inspection plan. Normally, this field remains empty, as in nearly every case only the operation number is used in inspection planning. The operation designation is only used in exceptional cases.

Quantities**Rework**

A rework quantity may be entered in this field when the inspection step is completed. This quantity specification is used for informative purposes only and is not processed by default. Normally, this field is only used if the shop floor data collection module (BDE) is not used in production, i.e. only the CAQ module is in use. Moreover, this field is useful in the goods receipt and goods issue area.

Scrap

A scrap quantity may be entered in this field when the inspection step is completed. This quantity specification is used for informative purposes only and is not processed by default. Normally, this field is only used if the shop floor data collection module (BDE) is not used in production, i.e. only the CAQ module is in use. Moreover, this field is useful in the goods receipt and goods issue area.

Actually produced quantity (act. prod. qty.)

The actually produced quantity may be entered in this field when the inspection step is completed. This quantity specification is used for informative purposes only and is not processed by default. Normally, this field is only used if the shop floor data collection module (BDE) is not used in production, i.e. only the CAQ module is in use. Moreover, this field is useful in the goods receipt and goods issue area.

Inspection step - editing functions

An inspection step cannot be edited. Only quantity specifications may be added while completing inspection steps.

Inspection step - toolbar**Release**

Function authorization: irisp.release

Transfers an inspection step of the status "created" or "completed" to the "released" status or to the status that existed at the time when it was completed, e.g. to the "partial result" status.

**Complete**

Function authorization: irisp.complete

Opens a dialog where the (inspection) result is preset on the basis of the existing inspection results of the characteristics. By completing the inspection step, it is finally transferred to the "completed" status. Inspections can no longer be performed for inspection steps in this status.

**Cancel**

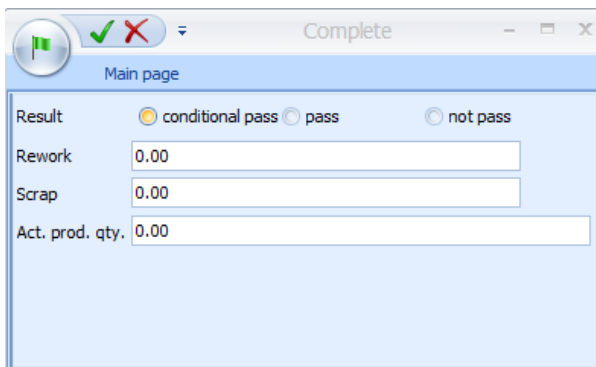
Function authorization: irisp.cancel

Transfers the inspection step to the "canceled" status. Inspections can no longer be performed for inspection steps in this status.

"Complete inspection step" detail application

Function authorization	irisp.complete
------------------------	----------------

The corresponding button has to be clicked in the toolbar to complete an inspection step. The following editing dialog opens afterwards.



The system suggests the result "pass" if all corresponding inspection step characteristics have the inspection result "pass" or if they are not yet checked. The system suggests the result "fail" if there is at least one inspection step characteristic that has the inspection result "fail". However, the user is free to choose another result.

The "rework quantity", "scrap quantity" and the "actually produced quantity" may be entered when the inspection step is completed (irrelevant to gage management). This quantity specification is used for informative purposes only and is not processed by default. Normally, this field is only used if the shop floor data collection module (BDE) is not used in production, i.e. only the CAQ module is in use. Moreover, these fields are useful in the goods receipt and goods issue area. In particular in the goods receipt area, it is useful to record scrap and rework as these details may be uploaded for several inspection steps to a higher-level system at a later point in time when inspection requirements are completed.

"Inspection points" detail application

The standard configuration of the system specifies that inspection points are only used in the production area. Using inspection points only makes sense if several samples are drawn and checked within an order/inspection requirement. This is, for example, the case if inspections are to be performed subject to different events in production. This might, for example, be the case if a specified inspection interval based on quantities or time is reached. As the individual inspection plan characteristics might have different events that make inspections become due, the inspection points of an inspection step might indeed have different inspection characteristics.

Once an inspection has been performed, every inspection point is to be completed. Every inspection point has an inspection result (pass, fail) and a status (completed, open). Inspection points that have not yet been checked have a "pass" inspection result by default.

Please note

If text exceeding the field lengths existing in the relevant inspection point dialog is entered in inspection point fields that are also visible/used on AIP terminals, this field content will be cut off when saving the inspection point in AIP. Therefore, the field contents should not be longer than the field lengths configured for the AIP inspection point dialog. The field lengths of AIP dialogs can be adjusted as part of system customization.

"Inspection points" field descriptions

The essential inspection point fields are described in the paragraphs that follow. Self-explanatory fields are not explained.

Inspection point**Inspection point number**

The inspection point number identifies the inspection point uniquely among the inspection points pertaining to the inspection step. The superordinate key fields "inspection step number" and "inspection request number" are shown in addition to the "inspection point" field.

Inspection result

The inspection result can be "pass" or "fail". Inspection points that have not yet been checked have the inspection result "pass" by default (standard system configuration).

Status

An inspection point can be completed or open. Only open inspection points may still be checked at the AIP terminal.

Inspection point identification

The fields, which have been identified as being activated in the superordinate inspection step, are shown here. The CAQ system options define which fields identify the inspection point and which ones require input. The following fields are displayed by default, whereas none of these fields requires input.

Date

Allows for a date to be entered. The system date is preset in case new inspection points are created. This date may directly be changed if they are created manually. Otherwise, the editing mode allows for changes to be made.

Time

Allows for a time to be entered. The system time is preset in case new inspection points are created. This time may directly be changed if they are created manually. Otherwise, the editing mode allows for changes to be made.

In addition to the configurable identification fields, there are still the following identification fields that are always available.

Workplace

If the inspection point is generated at an AIP terminal or the automatic generation is triggered by an inspection event of a specific terminal, this field shows the workplace of the triggering inspection station. If an inspection station is specified this inspection point will only be available at this inspection station.

Batch

Possibility to enter a corresponding batch.

Partial batch

Possibility to enter a corresponding partial batch.

Details**Generation of inspection points**

If an inspection point is generated automatically on the basis of reaching a defined time or quantity interval, it has the flag "relating to time" or "relating to quantities". An inspection point that is assigned the flag "free" has been generated manually or on the basis of other events that make inspections become due.

Inspection points - editing functions

New inspection points cannot be created for completed or canceled inspection steps. The same applies to the modification of existing inspection points. To allow for an inspection point to be created or edited, the inspection step has to be released beforehand.

Inspection points - toolbar**Release**

Function authorization: iripp.release

Transfers a completed inspection point to the "open" status.

**Complete**

Function authorization: iripp.complete

Opens a dialog window that allows for inspection points to be completed.

"Complete inspection point" detail application

Function authorization	iripp.complete
------------------------	----------------

The corresponding button has to be clicked in the toolbar to complete an inspection point. Then the following editing dialog opens including an inspection result that cannot be changed and that is calculated on the basis of the inspection results for characteristics. In addition to this, in the tab "Evaluation" a usage decision for inspection points calculated on the basis of the inspection results for characteristics is preset, which may, however, be changed manually. To do so, there is a selection list of usage decisions for inspection points, filtered by the plant, whereas every usage decision finally classifies the inspection point either as "pass" or "fail", irrespective of the inspection result. Additional information is displayed on the usage decision selected for the inspection point, once the selection has been confirmed. Additional information may, for example, be the code group, code number, catalog type and the factory/site.

MPDV services for customizing the system may provide further usage decisions for inspection points.

According to the configuration of identification fields, they can still be edited when inspection points are completed. If some identification fields require input (mandatory fields), they need to be filled out before inspection points can be completed. In addition to this, quantity fields may also be filled out.

"Inspection step/inspection point characteristics" detail application

Function authorization	iriscp.*
------------------------	----------

The detail application of inspection steps/inspection point characteristics is nearly identical to the master data of characteristics application. For this reason, reference is made only to additional features in the following.



Go to

For further information on the definition of characteristics, please refer to the functional description of the document entitled [MOC_CharacteristicsQM](#).

The inspection step characteristics are displayed below the inspection step within the master-detail grid. Provided that inspection points are in use and inspection points have been generated, the list of inspection points may be opened below the inspection steps. This list again allows for the characteristics pertaining to the selected inspection point to be opened.

The existing inspection plan determines which characteristics are included in the inspection step. In case the inspection plan is configured in a way specifying that a separate inspection step is to be generated for each operation (applying to several workplaces), the inspection step only includes the characteristics of a specific operation. Provided that the inspection plan has additionally been configured to generate a separate inspection step for every workstation, an inspection step always includes the characteristics of a fixed combination of operation and workplace.

An inspection point, in turn, only includes characteristics of the same event causing inspections to become due. Even the manual generation of inspection points represents a due date event. An inspection point is generated automatically, once e.g. a quantity interval defined for the inspection step characteristics has been achieved. This inspection point includes all characteristics that have the same quantity interval or that have a quantity interval that has the achieved quantity interval as least common multiple.

Example:

Characteristic 1 has a quantity interval of 100 and characteristic 2 of 200. Once the first 100 items have been produced, an inspection point is generated, which only includes characteristic 1. After another 100 items that have been produced, another inspection point is generated that includes the characteristics 1 and 2.

If an inspection becomes due because of changing the machine status, the generated inspection point only includes characteristics that have to be checked on the basis of the same machine status change.

Only manually generated inspection points are not limited with respect to the corresponding characteristics. These inspection points always include all characteristics of the related inspection step.

Inspection step/inspection point characteristics field descriptions

The paragraphs that follow only describe the fields that are available in addition to the characteristic master data or inspection plan characteristics.

Characteristic

Initial sample creation

Shows the field content transferred from the basic inspection plan characteristic. A separate sheet/page is printed for each category in the printed initial sample report.

Properties

Dynamically modified

This checkbox is enabled, provided that a characteristic is dynamically modified.

Dynamic modification

Determined inspection severity/inspection severity designation

This field shows the inspection severity determined for this characteristic on the basis of the dynamic modification rules specified within the inspection plan.

Revised inspection severity/inspection severity designation

This field only exists if dynamic modification is based on characteristics. A new initial inspection severity is displayed here, provided that the user has intervened in the process of dynamic modification and changed the current inspection severity (new initial inspection severity).

Please note

A note is displayed in this field, in case an error has occurred with the dynamic modification of this characteristic while inspection steps are generated.

Details

No cavity

The authorization "ipl_cav" is required for displaying these fields.

This option allows for a characteristic to be highlighted as irrelevant to cavities, although the respective inspection requirement specifies that the relevant characteristics are (actually) to be recorded in relation to cavities.

This checkbox has always to be enabled for attributive characteristics and inspection charts.

Sample group

This field specifies which sample group this characteristic belongs to.

Sampling

If a characteristic is assigned a sample group, this checkbox specifies whether or not a sample is to be generated by way of this characteristic. If this is the case, it is a "sampling characteristic". Consequently, the checkbox is to be enabled. A sample group is to be assigned to the sampling characteristic.



If these fields are not available, you require a new program version of this application.

Specifications

Sample size

This field shows the determined sample size, if within an inspection plan characteristic a sampling scheme is defined that does not require the sample size to be indicated, as it is calculated on the basis of the actual quantity of the inspection requirement. This is, for example, the case for the sampling schemes "batch inspection" or "100% inspection".



Go to

The other fields of the respective tabs correspond to those of the characteristic master data and are described in the documentation entitled "[CAQ characteristic master data](#)".

Inspection step/inspection point characteristics - editing functions

Inspection step or inspection point characteristics cannot be edited.

"Inspection request documents" and "characteristic documents" detail applications

Order	Id	Article designation	Drawing issue nu...	Drawing number	Usage decision
12151133	77 9938 882	Metal Pen, 4-colour, 10000m capacity	C	23213 90939	
QM987200	SAL-FFI-9005	mirror left black		821387663	

Position	Designation	Type	File name/...	Display	HYDRA pat...	Display UR...	Created	Created by
1	mirror	file	mirror.bmp	☒	QM_FILE	mirror.bmp	15/02/2010	HYDRA
2	working inst...	file	working_inst...	☐	QM_FILE	working_inst...	16/02/2010	Miller

The above screenshot shows how an inspection requirement document is assigned.

Provided that the "documents" tab has been activated, as many documents as required may be assigned to each inspection requirement and each inspection step characteristic within the master-detail grid. If this tab is activated in the master-detail grid corresponding editing buttons are enabled in the toolbar to edit the documents.

When documents are assigned, all formats registered by Windows are provided. Consequently, it is possible to assign simple documents (e.g. written in Word), drawings of any format and videos. However, the corresponding programs that are able to display the required formats have to be installed. In this context, the documents are opened by the program that has been linked in Windows.

The file types are "File", "URL" and "Text". The file name including the path may be entered manually for the "file" type. The "URL" file type enables access to the Internet or Intranet. The third file type "Text" allows for text to be entered directly.

A designation may be assigned to each defined document. Moreover, it may be determined in which order the documents are to be listed. The "position" field is used for this purpose (numeric input). The specifications made within this list must be unique. In addition, the checkbox "display with inspection" determines whether or not the document may be shown during the inspection process.

"Inspection plan documents" and "documents for inspection plan characteristics" toolbar

In addition to the standard functions, there is also the button to show the documents.



Show documents

If a document link is defined this button opens and shows this document. However, a program, which can show the linked file type, has to be installed on the PC.

"Failure" detail application

A "failure list" detail application is provided on the level of inspection requirements. This list shows all failures of the type "failure type", "failure location", "failure cause" and "originator", which have been assigned to the characteristics, samples or measured values during the inspection process. The list even shows the failure types that have been generated automatically, e.g. "upper tolerance limit violated".

In addition to the respective failure, the following referenced data is also available.

- Inspection step
- Operation sequence number (OP sequence, AFO)
- Sample number
- Value number
- Failure type
- Characteristic number and designation
- Workplace number and designation
- Weighting (number, e.g. for an inspection chart characteristic)
- Comment,
- Failure date,
- Failure time

The list may be restricted in a flexible way by using the individual column filter. In addition, the grouping function allows for failures to be listed for every characteristic, for example.

Failure list - editing functions

Failure analysis entries can be created, edited or deleted for inspection requirements. If a new data record is created, further key information, such as

- Inspection step number,
- OP sequence number (uniquely identifies the characteristic),
- Sample number
- Value number

may be added to the assignment.

The target group can be found in the administrative area, as "administrative" skills are required for more precise assignment of this key information.

These authorizations are required.

Function authorization	failure.create => create new failure failure.edit => edit failure failure.delete => delete failure
------------------------	--

Users must have profound knowledge of the inspection processes as they are not guided through the process of data collection/modification and validation checking is not performed.

The fields "area" and "inspection requirement number" can neither be modified nor assigned values during initial data creation. This data is taken from the previously selected inspection requirement when it comes to initial data creation.

The other fields can be assigned values as part of initial data creation. Only the fields "weighting" and "comment" may be changed in the editing mode.

"Measures" detail application

A "measure list" detail application is provided on the level of inspection requirements. This list shows all measures which have been assigned to the characteristics, samples or measured values during the inspection process.

In addition to the respective measure number and designation, the following data is also referenced.

- Inspection requirement
- Inspection step
- Operation sequence number (OP sequence, AFO)
- Sample number
- Value number

- Measure type
- Party in charge incl. type
- Status
- Comment
- Text
- Effectiveness

The list may be restricted in a flexible manner by using the individual column filter. The grouping function provides further analysis options.

"Measures" toolbar

The button "measure tracking" can be used to open the measure tracking dialog to edit and create measures. Measure tracking is filtered by the internal and unique "serial number" of the measure in the "references" tab. Filtering is omitted if no measure is selected.



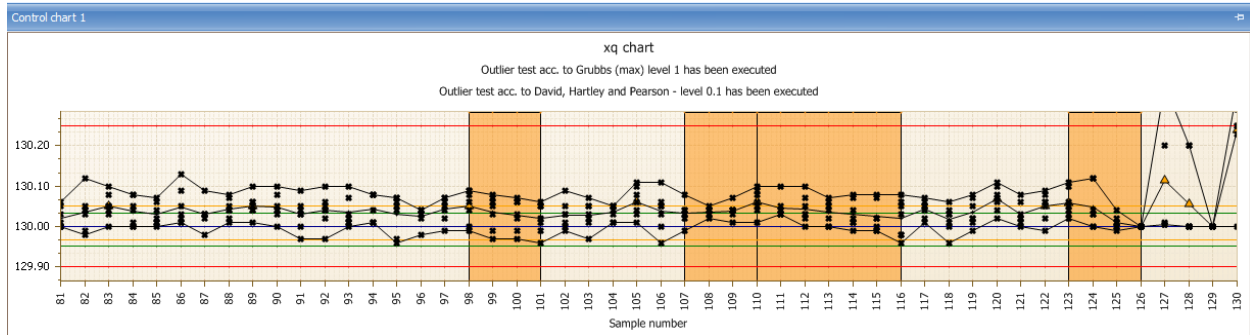
Go to measure tracking

Function authorization	failure.* => opens measure tracking
------------------------	-------------------------------------

"Control chart 1 / 2" detail application

The control charts 1 and 2 defined for the inspection step characteristic are displayed by default. In case they are not defined for the inspection step characteristic, the xq and s charts are used for variable characteristics and the p and u chart for attributive characteristics.

All data of the selected inspection step characteristic is used as basis for graphic preparation.



The contents of this application are defined by opening the dialog to configure "control chart 1" or "control chart 2". Changes made via this dialog are saved according to the user's requirements. Provided that no special settings have been made, the xq chart ("variable" characteristic type) or the p chart ("inspection chart" or "attributive" characteristic type) is displayed, by default. Subject to the characteristic type, the user may choose to display one of the following control charts by default.

Variable characteristic

- Xq chart
- s chart
- R chart
- Single value chart
- Median chart

Attributive characteristic

- p chart
- np chart
- c chart
- u chart

mpciv SettingsDialog

Chart type:

Number of samples/measured values to be requested: ☐ All ☒ Quantity

☒ Show trend ☒ Show information on outlier tests

☒ Show run ☒ Perform outlier test according to Grubbs max. (1%)

☐ Show middle third ☐ Perform outlier test according to Grubbs max. (5%)

☒ Show single values ☐ Perform outlier test according to Grubbs min. (1%)

☒ Show target value ☐ Perform outlier test according to Grubbs min. (5%)

☒ Show tolerance limits ☒ Perform outlier test according to David-Hearty-Pearson (0.5%)

☒ Show action limits ☐ Perform outlier test according to David-Hearty-Pearson (1%)

☒ Show warning limits ☐ Perform outlier test according to David-Hearty-Pearson (5%)

☒ Show mean value

☐ Consider long-term data

☒ Show title of current label Control chart labeling:

☒ Consider number of decimal places Number of decimal places:

☐ Show characteristic title

☒ Show control chart type

☒ Combine minimum and maximum values ☒ Show legend of y-axis

☒ Show grid ☐ Automatic scaling

The paragraphs that follow explain the essential configuration options.

Number of the samples to be requested

Specifies how many samples or single measured values are to be displayed in the control chart.

Display ...

The options that include the term "display ..." enable or disable the presentation of the corresponding data in the control chart.

Consider long-term data

This option has to be checked to integrate archived data from the medium-term data area.

Combine minimum and maximum values

It is recommendable to show the minimum and maximum values that are each connected by a line to improve the presentation of the range of dispersion of single values.

Automatic scaling

The "automatic scaling" function allows for all values to be displayed, irrespective of the existing limit values. This shows even extreme outliers in the control chart. The disadvantage is that there is less space for the other measured values and, it is very often the case that a changing value pattern can hardly be recognized.

Show trend / run / middle third

The monitoring functions "trend", "run" and "MiddleThird" allow better surveillance of a process than by using the control chart alone. However, they can only be used with control charts for xq and median. The trend allows visualization of an upward or downward tendency in the process over several samples. By default, these are seven subsequent rising or falling values. The run shows sections in which the process runs above or below the mean value (when displayed, otherwise the target value) over several samples. By default a run is recognized if seven subsequent values are above the mean value. The number of seven samples/values for recognizing a trend or run, which is set by default, may be changed by customizing the system. "MiddleThird" refers to an unusually high or low number of values, in the section of the control chart viewed, within the middle third of the area bounded by the action limits.

Respective analyses are performed automatically for "trend", "run" and "MiddleThird". The control chart shows in a graphic if a trend, run and / or MiddleThird is available/detected. The following events are altogether represented by icons or color codes in the control chart.

- Trend (can be recognized by a colored area)
- Run (can be recognized by a colored area))
- Middle Third (can be recognized by a colored area)
- Outlier (icon: )
- Xq violates action limit (icon )

The presentation of outliers is restricted to the xq chart and median chart and connected with the presentation of single values. There are different functions for performing and presenting outlier tests. The different outlier tests for the different levels may be activated separately. Provided that the presentation of outlier tests has been activated, the result of this test is displayed in text form above the corresponding control chart.

The below outlier tests are available for the different inspection levels, whereas the inspection level is indicated in parentheses.

- Grubbs max. (1 %)
- Grubbs max. (5 %)
- Grubbs min. (1 %)
- Grubbs min. (5 %)
- David-Hartley-Pearson (0.5 %)
- David-Hartley-Pearson (1 %)
- David-Hartley-Pearson (5 %)

Notes on outlier tests

- The outlier tests do not refer to the collectivity of all samples, i.e. every sample is considered individually. Consequently, the following phenomena may appear:
 - Despite a large range between minimum and maximum value no outlier can be identified. A reason for this may be the equal distribution of the single values within the sample.
 - Despite a low range between minimum and maximum value an outlier can be identified. A reason for this may be an accumulation of many values at one "point" so that an individual value having a certain distance to this agglomeration is identified as outlier within the sample.
- At least three values have to be in the sample in order to be able to perform an outlier test. The more values are available within a sample the more uniform the general view of the outliers becomes compared to all samples.
- The Grubbs outlier test is performed with a sample size of $2 < n < 148$ (n = number of values within a sample). The outlier test according to David, Hartley und Pearson is performed at a sample size of $2 < n < 1251$ (n = number of values within a sample).

X-axis labeling

The below information is available to label the x-axis of control charts.

- Sample number
- Order number
- PPS reference number
- Purchase order number
- Batch
- Article number
- Machine number
- Cavity number
- Date + time of the first measured value of a sample
- Date + time of the last measured value of the sample
- Date + time of sample completion
- Badge number of the first measured value of the sample
- Badge number of the last measured value of the sample
- Badge number of sample completion

If these fields are not available, you require a new program version of this application.



- Partial batch
- Workplace
- Production workplace

- Field 1
- Field 2
- Field 3
- Field 4
- Field 5
- Field 6
- Field 7
- Field 8



The fields "field 1" to "field 8" are enabled subject to their usage by MPDV customizing and assigned an individual designation. As these field names are flexible, they are only entitled "field 1" to "field 8" in this document. Field 1 includes, for example, the tool if cavity-related data collection is enabled.

Tool tips within the control chart

When a value is labeled being an outlier (red rhomb) detailed information on which test(s) was (were) the crucial factor for this determination is displayed when going with the mouse over this labeling (rhomb).

For mean values the tool tip shows the value, date and time for the first and last measured value of this sample as well as the corresponding inspection request number and inspection step number.

The exact value is shown for single measured values.

A special note indicating why an area is colored is also displayed for the colored areas when a trend, run or middlethird is recognized.

Sorting of measured values

The type of control chart sorting is determined by the server and depends on how the control chart filters are configured. If the filters "operation sequence" and "inspection request number" or "operation sequence" and "inspection step number" are indicated the characteristic is identified uniquely and sorting is based on the sample number.

Sorting is based on date and time if either the filter field "operation sequence" is left empty or it is filled out and the fields "inspection request number" and "inspection step" number are left empty.

The optical presentation of trend or run always refers to how the measured values are sorted. However, the automatic generation of the failure type "trend" is always based on the measured values being sorted by the sample number, as the numbers for the operation sequence, inspection request and the inspection step are always known at the time data is recorded.

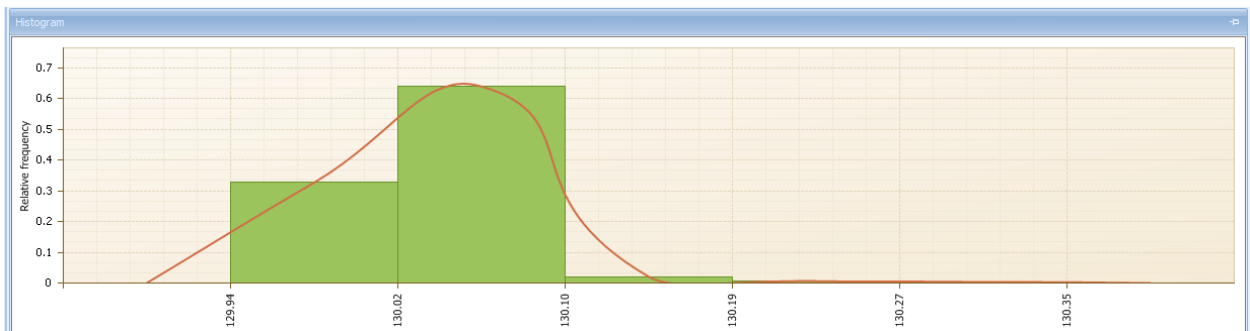
Consequently, the following scenario might be possible.

- The control chart is sorted by date and time and a trend is shown, although the automatic failure "trend" has not been generated.

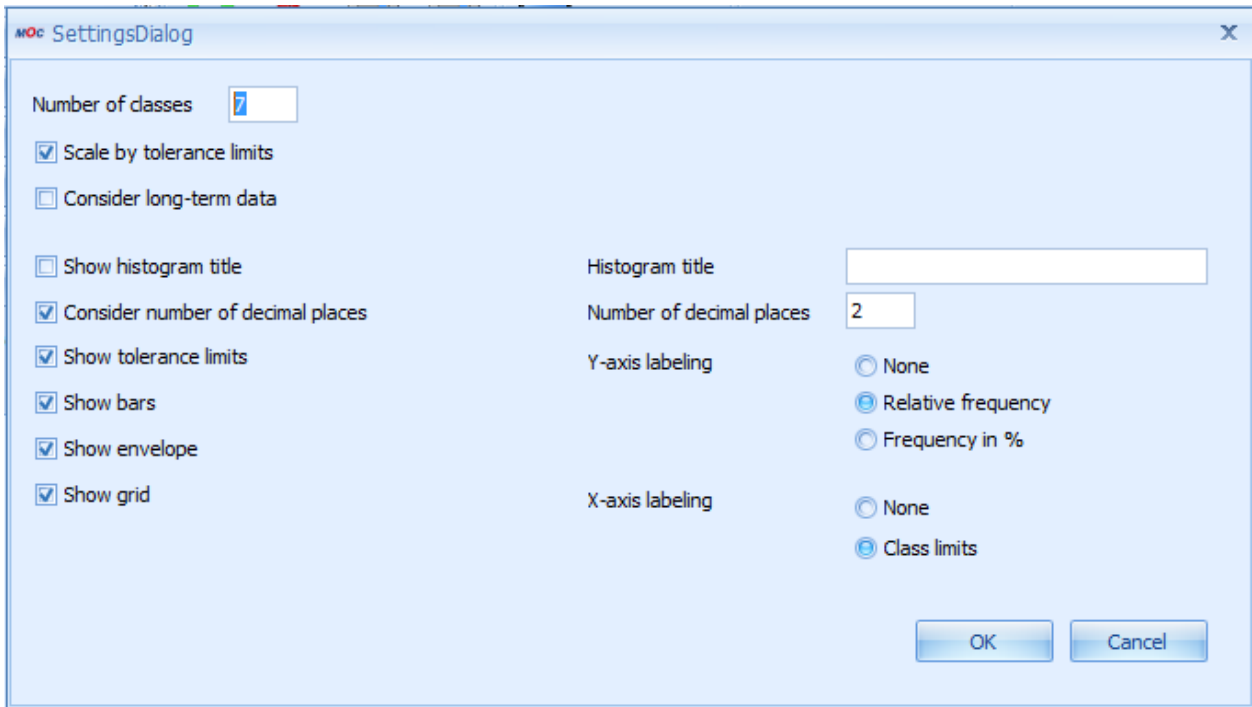
The control chart is sorted by the sample number and no trend is shown, although the automatic failure "trend" has been generated.

"Histogram" detail application

All data of the selected inspection step characteristic is used as basis for graphic preparation.



Unlike the control charts, which display a specified excerpt from the sample set, the histogram is always based on the entire set of available samples matching the selection filter criteria. The appearance of the histogram is determined by the number of classes and by any elements additionally displayed. The contents of this application are defined by opening the dialog to configure the "histogram". Changes made via this dialog are saved according to the user's requirements.



The paragraphs that follow explain the essential configuration options.

Number of classes

Determines the number of histogram classes according to which the measured values are to be distributed. If represented within the tolerance limits (option "scale by tolerance limits") one histogram class each is outside of the tolerance limits.

Scale by tolerance limits

Enabled: The classes are in between the tolerance limits with each one "outlier class" to the left and to the right.

Disabled: The classes include the range of all measured values (no separate classes for values outside of the tolerance limits).

Consider long-term data

Includes the archived data from the medium-term data area.

Show histogram title

If a special title is to be displayed it may be entered by enabling this option.

x-axis labeling

The x-axis shows the corresponding values of the class limits if the "class limits" option is set. The number of decimal places that are to be displayed may be set by the two configuration options for decimal places.

Consider the number of decimal places

Takes into account the defined number of decimal places in the x-axis labeling.

Control chart 1/2 and histogram

Control chart 1 settings

Opens a dialog to configure the settings of control chart 1. The corresponding details are described in the respective detail application.

Control chart 2 settings

Opens a dialog to configure the settings of control chart 2. The corresponding details are described in the respective detail application.

Histogram settings

Opens a dialog to configure the histogram settings. The corresponding details are described in the respective detail application.

"Samples" detail application

The statistical key figures about every single sample can be found in the "samples" detail application on the level of inspection step characteristics. In addition to the referenced key fields, such as the inspection requirement number, inspection step number and sample number, provided that they are available or can be calculated, the key figures

- Xq
- Xq floating
- R floating
- Standard deviation s
- s floating
- Minimum
- Maximum
- Range
- Median
- P
- U
- Number of defects

are listed. The list additionally shows the characteristic specifications (tolerance, action and warning limits) as well as the referenced machine.

"Samples" toolbar



Complete sample

It might be necessary to complete samples manually, as measured values can be entered on "administration" level or as there might be inspections without inspection points including the creation of new samples and/or inspections without having reached the sample size. Samples can even be completed if they have not been completed by AIP inspection data collection.



Release sample

This function enables samples to be released once more and the collection of additional measured values for this sample number, provided the defined sample size has not yet been reached.

"Single values" detail application

The single measured values for all variable inspection step characteristics can be found in the "single values" detail application on the level of inspection step characteristics. For attributive characteristics the following information

- Number of defects
- Number of NCU (non-conforming units) and
- Defects

is displayed. Moreover, it can be recognized whether the measured value or the attributive assessment is valid or invalid.

The characteristic specifications (tolerance, action and warning limits) are listed in addition to the referenced key fields, such as the inspection requirement number, inspection step number, sample number, and value number. Every entry also shows the date and time when it was recorded and edited as well as the responsible user.

Single values - editing functions

Measured values can be created, edited or deleted for a variable inspection step characteristic and/or inspection point characteristic and the number of non-conforming units may be created, edited or deleted for attributive characteristics on "administration" level. If a new data record is created, further key information, such as

- the sample number
- and value number

may be added to specify the assignment.

The target group can be found in the administrative area, as "administrative" skills are required to provide for more precise assignment of this key information. Users must have profound knowledge of the inspection processes, as they are not guided through the process of data collection/modification and validation checking is not performed.

These authorizations are required.

Function authorization	value.create => create new measured value value.edit => edit measured value value.delete => delete measured value
------------------------	---

Users must have profound knowledge of inspection processes, as they are not guided through the process of data collection/modification and validation checking is not performed.

The fields

- Area,
- Inspection request number,
- Inspection step no.,
- Inspection step number (only available for inspection point characteristics),
- OP sequence (uniquely identifies the characteristic),
- Upper tolerance limit,
- Target value,
- Lower tolerance limit and
- Number of decimal places

can neither be modified nor assigned values during initial data creation. When it comes to initial data creation, this data is taken from the previously selected characteristic and the referenced inspection requirement as well as from the inspection point.

The other fields can be assigned values and/or changed during initial data creation.

The field "single value of cavity" is only visible if

- the inspection requirement shows an ID indicating that data is generally collected in relation to cavities and
- the relevant characteristic is not assigned an ID indicating that it still has to be checked without relation to cavities.

If a new data record is created, the field for the measured value will be restricted to the number of decimal places. The other decimal places are filled up with zeros. For technical reasons, these "additional" decimal places are shown during processing.

The field "measured value" is shown for variable characteristics. Instead of the measured value field, the fields "sample size (checked)", and "number of DU" are shown for attributive characteristics, i.e. this also includes inspection chart characteristics.

"Statistics" detail application

The common statistical key figures about the selected characteristic can be found in the "statistic" detail application on the level of inspection step characteristics.

For variable characteristics these are

- Number of samples
- Number of measured values
- $\bar{x}$
- Minimum
- Maximum
- R
- S
- S relative
- Sigma
- Cp and
- Cpk

For attributive characteristics these are

- Number of non-conforming units

- Number of defects
- p and
- u .